

Pore-Scale Study of Transverse Mixing Induced CaCO_3 Precipitation and Permeability Reduction in a Model Subsurface Sedimentary System

CHANGYONG ZHANG,[†] KARL DEHOFF,[‡]
NANCY HESS,[†] MART OOSTROM,[§]
THOMAS W. WIETSMA,^{||}
ALBERT J. VALOCCHI,[‡]
BRUCE W. FOUKE,[⊥] AND
CHARLES J. WERTH^{*†‡}

Fundamental and Computational Sciences Directorate, Pacific Northwest National Laboratory, Richland, Washington 99352, Department of Civil and Environmental Engineering, University of Illinois at Urbana-Champaign, 205 N Mathews Ave, Urbana, Illinois 61801, Energy and Environment Directorate, Pacific Northwest National Laboratory, Richland, Washington 99354, Environmental Molecular Sciences Laboratory, Pacific Northwest National Laboratory, Richland, Washington 99354, and Department of Geology, University of Illinois at Urbana-Champaign, 1301 W Green St, Urbana, Illinois 61801

Received June 16, 2010. Revised manuscript received August 13, 2010. Accepted August 18, 2010.

A microfluidic pore structure etched into a silicon wafer was used as a two-dimensional model subsurface sedimentary system (i.e., micromodel) to study mineral precipitation and permeability reduction relevant to groundwater remediation and geological carbon sequestration. Solutions containing CaCl_2 and Na_2CO_3 at four different saturation states ($\Omega = [\text{Ca}^{2+}][\text{CO}_3^{2-}]/K_{\text{spCaCO}_3}$) were introduced through two separate inlets, and they mixed by diffusion transverse to the main flow direction along the center of the micromodel resulting in CaCO_3 precipitation. Precipitation rates increased and the total amount of precipitates decreased with increasing saturation state, and only vaterite and calcite crystals were formed (no aragonite). The relative amount of vaterite increased from 80% at the lowest saturation state ($\Omega_v = 2.8$ for vaterite) to 95% at the highest saturation state ($\Omega_v = 4.5$). Fluorescent tracer tests conducted before and after CaCO_3 precipitation indicate that pore spaces were occluded by CaCO_3 precipitates along the transverse mixing zone, thus substantially reducing porosity and permeability, and potentially limiting transformation from vaterite to the more stable calcite. The

results suggest that mineral precipitation along plume margins can decrease both reactant mixing during groundwater remediation, and injection and storage efficiency during CO_2 sequestration.

Introduction

Geological carbon sequestration is being evaluated at several field sites around the world to determine its technical feasibility, economical viability for storing carbon produced from stationary sources (e.g., refs 1–5) and the ability to mitigate climate change (6). Carbon dioxide capture, transport, and storage (CCS) (7) involves the capture of CO_2 from fossil fuel combustion, compression of captured CO_2 into a liquid or supercritical fluid state, transport and subsequent injection. The most volumetrically significant subsurface rock targets for injection are deep subsurface saline aquifers, depleted oil reservoirs, and unmineable coal seams (8). Injected CO_2 displaces brine or oil in pores and also rises vertically toward the caprock, being less buoyant than the saline brines housed within subsurface sedimentary reservoirs. Along fluid–fluid interfaces between injected CO_2 and pore water, CO_2 dissolves into the aqueous phase and either reduces the pH, or reacts with species that buffer pH to increase bicarbonate and carbonate concentrations (9, 10). Decreasing pH conditions induce carbonate mineral dissolution, which release divalent metal cations, and increase bicarbonate and carbonate concentrations. Simulation results indicate that these conditions promote carbonate mineral precipitation along the margins of injected CO_2 plumes (11–13). Two significant concerns associated with mineral dissolution and precipitation during CO_2 injection are dissolution and failure of the caprock seal, and precipitation of minerals that reduce permeability, reservoir capacity, and CO_2 injection efficiency.

A large body of literature has explored dissolution and precipitation reactions relevant to CO_2 injection in laboratory batch studies (e.g., refs 14–16). Which mineral species react and the time scales of reaction depend on a number of factors including existing minerals and solutes, pH, temperature, and pressure (17–19). Among the most common minerals available for reaction in geologic storage reservoirs are calcium carbonates, which include vaterite, aragonite, and calcite (20). They are often present in limestone in sedimentary deposits, and dissolution and precipitation time scales are rapid (i.e., minutes to days). Carter et al. (20) showed that in a batch system, vaterite is initially the major phase at low temperature (i.e., 20 °C), and aragonite is initially the major phase at temperatures above 50 °C; both transformed to the more stable form calcite after 7 h.

Although calcium carbonate dissolution and precipitation reactions have been extensively studied in batch systems (e.g., refs 9, 15, 16, and 20), very little work has explored these reactions under flow conditions, where advective and diffusive mixing processes play a major role (21, 22). Preliminary modeling studies (11–13) indicate that along the fringes of injected CO_2 plumes, convective and diffusive mixing processes affect CO_2 dissolution into the pore water and subsequent dissolution and precipitation reactions within the formation and caprock. Calcium carbonate precipitation can decrease porosity and permeability. This has been shown in wellborn cement and caprock core samples (23), and in controlled two-dimensional (2D) benchtop flow cell experiments where calcium carbonate was precipitated along the center mixing line between aqueous phase calcium and carbonate (21, 22). Calcium

* Corresponding author e-mail: werth@illinois.edu; phone: 217-333-3822; fax: 217-333-6968.

[†] Fundamental and Computational Sciences Directorate, Pacific Northwest National Laboratory.

[‡] Department of Civil and Environmental Engineering, University of Illinois at Urbana-Champaign.

[§] Energy and Environment Directorate, Pacific Northwest National Laboratory.

^{||} Environmental Molecular Sciences Laboratory, Pacific Northwest National Laboratory.

[⊥] Department of Geology, University of Illinois at Urbana-Champaign.

TABLE 1. Summary of Experimental Conditions

exp	conc. [mM]	CaCl ₂ pH ^a	Na ₂ CO ₃ pH ^a	Ω _c wrt calcite ^b	Ω _a wrt aragonite ^c	Ω _v wrt vaterite ^d
1	6.5	5.8	10.6	3.4	3.2	2.8
2	10	5.9	10.8	3.8	3.6	3.1
3	25	6.1	10.9	4.6	4.4	3.9
4	50	6.3	11.1	5.2	5.0	4.5

^a pH of DI water used to prepare all solutions is 5.7. ^b $pK_{sp} = 8.42$. ^c $pK_{sp} = 8.22$. ^d $pK_{sp} = 7.73$; all pK_{sp} values from Stumm (36).

carbonate precipitation limited by advective or diffusive mixing is relevant to other subsurface processes as well, including precipitation near injection wells and along plume margins during groundwater remediation (24), precipitation along fluid injection fronts during enhanced oil recovery (25), and precipitation of leachate along the shadow zones of nuclear waste storage tunnels (26).

Herein, we focus on calcium carbonate precipitation and dissolution under transverse mixing limited conditions. Reactive transport limited by transverse mixing has been the focus of a number of recent studies. However, in all but a few cases reactions of interests were limited to soluble reaction products (e.g., refs 27–30). Transverse mixing-limited reactive transport involving mineral precipitation and dissolution has been limited to numerical simulations (e.g., refs 21, 22, and 31–34) and two studies in continuum-scale flow cells (e.g., refs 21 and 22). These studies all indicate that mixing limitations between dissolved cations and carbonate give rise to highly nonuniform concentration gradients, localized reactions and precipitation, and hence highly localized changes in porosity and permeability. However, experimental pore-scale studies are lacking, and the influence of these conditions on mineral growth rate, morphology, phases, and pore blockage are unknown.

The objective of this study is to investigate calcium carbonate precipitation limited by transverse mixing between calcium chloride and sodium carbonate in the context of flow through porous subsurface sedimentary systems. Specifically, we evaluate the impact of transverse concentration gradients and solution saturation state on calcium carbonate precipitate morphology, phase, stability, and reaction rate in a homogeneous pore network etched into a silicon substrate (i.e., micromodel). Reflected differential interference contrast microscopy is used to evaluate the precipitate morphology, stability, and reaction rate, and confocal Raman spectroscopy is used to evaluate the precipitate phase. Micromodel flow and transport systems are excellent platforms for elucidating coupled processes involving advection, diffusion, and biogeochemical reactions at the pore-scale, and for providing insights to continuum-scale processes (30, 35). While reactant species other than calcium and carbonate are also important in subsurface sedimentary systems, this research focuses on these as the first step in evaluation of coupled pore-scale physical processes (advection, diffusion) and geochemical reactions that potentially impact CO₂ sequestration and other subsurface processes.

Materials and Methods

Chemicals. Solutions of calcium chloride (CaCl₂), sodium carbonate (Na₂CO₃), and hydrochloric acid (HCl) (all chemicals are ACS reagents grade, Sigma-Aldrich) were prepared in deionized (DI) water and degassed under sonication prior to use. Four equal molar concentrations of CaCl₂ and Na₂CO₃ were prepared: 6.5, 10, 25, and 50 mM. Table 1 lists the pH and saturation state (Ω) of each set of solutions with respect to (wrt) three different polymorphs of CaCO₃ (i.e., calcite, aragonite, and vaterite) (36). The saturation state is defined



FIGURE 1. Micromodel and flow setup.

as $\Omega = [\text{Ca}^{2+}] \times [\text{CO}_3^{2-}] / K_{sp\text{CaCO}_3}$ (37); concentrations of Ca²⁺ and CO₃²⁻ are based on the inlet reactant concentrations. The experimental conditions were chosen as a first step to evaluate mineral precipitation rates, morphology, phase, and the corresponding effects on permeability; they do not directly reflect the range expected in an actual reservoir.

Fluorescein (Sigma, 70%) was dissolved in DI water at 10 μM and used as a fluorescent tracer to visualize changes in permeability as a result of CaCO₃ precipitation.

Micromodel Experiments. The micromodel (Figure 1) used in this study consists of two separate inlet channels (A, B) connected to a 2 cm × 1 cm (length × width) pore network, followed by a third inlet (C) and an outlet (D). The pore network contains a uniform distribution of cylindrical posts 300 μm in diameter, with 180 μm pore spaces, 40 μm pore throats, and a porosity of 0.39. The depth of the pore network is 17–21 μm. Standard photolithography techniques and ICP-DRIE (inductively coupled plasma-deep reactive ion etching) were used to fabricate the micromodels; details of the microfabrication process can be found in Chomsurin and Werth (38) and Willingham et al. (29). The fabricated silicon micromodels were heated to 1100 °C under oxygen for 1 h to grow a thin oxide layer (~0.1 μm). The final step was to anodically bond a cover sheet of Pyrex glass to etched silicon micromodel to seal the flow channels.

Prior to each experiment, the micromodel was first flushed with CO₂ gas to facilitate subsequent saturation by water, followed by >100 pore volumes (PV) of degassed DI water. Solutions of CaCl₂ and Na₂CO₃ were introduced separately into the pore network via inlets A and B at equal flow rates corresponding to a Darcy velocity of 1.25 cm/min; constant flow through the micromodel was maintained using a pair of ISCO pumps (Teledyne ISCO Inc., model 100DM). To prevent CaCO₃ precipitation and clogging in the outlet tubing, a solution of HCl (pH = 1.5–2.0) was introduced through inlet C using a syringe pump (Harvard Instruments) at a volumetric flow rate equal to one-half of the total flow rate through the pore network. An independent test using a fluorescent tracer showed minimal flow disruption in the pore network due to injection through C. All experiments were performed in duplicate, and with previously unused micromodels.

Microscope Imaging. Mixing between the two reactant solutions resulted in calcium carbonate precipitation in the center of the micromodel, which was imaged at selected time intervals until precipitation reached pseudo steady state (i.e., the number of pixels occupied by CaCO₃ precipitates in a micromodel was constant over time). All micromodel images were acquired with a Nikon Eclipse 2000 TE epifluorescent microscope using 4× and 10× inverted objectives (with image resolutions of 1.62 and 0.65 μm, respectively) and reflected differential interference contrast (DIC) microscopy. The microscope is equipped with an automated stage (Prior Scientific Instrument) and a monochrome digital Charge-Coupled Device (CCD) camera, both connected to a computer and controlled by NIS-Elements imaging software. A total of 75 (25 × 3) separate images were taken at each time point; an additional image from outside the pore network area was also acquired and applied to each image to correct for nonuniform illumination. Next, the 25 × 3 separate images

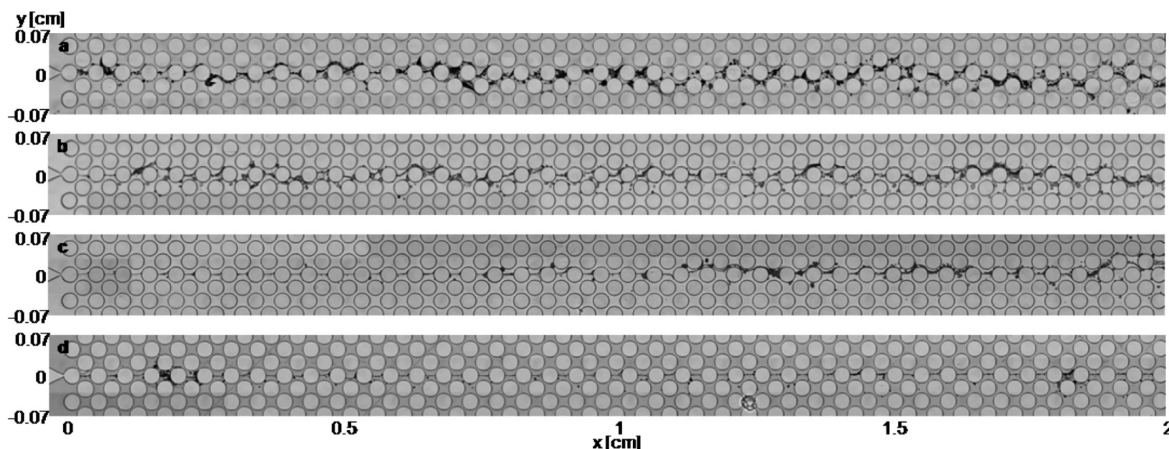


FIGURE 2. Images of center of micromodels with CaCO_3 precipitates formed along the mixing zone at different saturation states (a) $\Omega_c/\Omega_v = 3.4/2.8$, (b) $\Omega_c/\Omega_v = 3.8/3.1$, (c) $\Omega_c/\Omega_v = 4.6/3.9$, and (d) $\Omega_c/\Omega_v = 5.2/4.5$.

were montaged to form a single image, which captured the center of the pore network where CaCO_3 precipitates were formed.

Image Analysis. Montaged images ($0.65 \mu\text{m}$ resolution) of micromodels were analyzed to quantitatively evaluate CaCO_3 precipitation rates. The image intensity of CaCO_3 precipitates is lower than that for silicon posts and water-filled pore spaces; a threshold value was determined for each image to distinguish pixels with CaCO_3 precipitates, and the area (i.e., number of pixels) corresponding to precipitates was determined. This method only results in an apparent indicator of precipitation rate, since precipitates may not occupy the entire depth of the micromodel, and because the density of precipitates may vary between pixels.

Raman Back-Scattering Spectroscopy. This method was used to characterize calcium carbonate mineral phases in selected pores along the mixing zone when precipitation in a micromodel reached pseudo steady state. A long working distance $50\times$ objective was used to locate precipitates in the micromodel, to help focus 30 mW of 514.0 nm polarized excitation from a CW Nd:YAG diode laser to an approximate $1.5 \mu\text{m}$ spot size and to collect the Raman scattered light from the precipitate. Video imaging of the sample under laser illumination allowed precise positioning of the focused laser beam. The backscattered Raman light was collected and focused at the $200 \mu\text{m}$ entrance slit of a JY 800 triple spectrometer with 800 g/mm gratings. The exit slit was maintained at $200 \mu\text{m}$ and the scattered light was then dispersed on to a liquid nitrogen cooled charged coupled device Ge detector. The detector was calibrated using Hg emission lines. Since the Raman spectrum of CaCO_3 polymorphs ranges from about 100 to 1600 cm^{-1} , collection of the entire spectrum in two segments is necessary. The Raman signal was averaged for a minimum of 90 s for each segment. Multiple Raman spectra were collected on the three morphologically distinct crystallite types, and the peak positions and lineshapes of the resulting spectra were analyzed using a commercially available software package, GRAMS/32. Since Raman spectra reflect the molecular symmetry of the crystalline lattice, the pattern of peaks can be used to distinguish between polymorphs of CaCO_3 (20). The collected Raman spectra were assigned to either vaterite or calcite crystal structure by comparison to both a spectral database and spectra reported in literature (20, 39, 40).

Fluorescent Tracer Experiments. Immediately after CaCO_3 precipitation reached pseudo steady state in a micromodel, a fluorescein tracer solution and water were separately injected into the micromodel through inlets A and B, at a 2:1 flow rate ratio. A separate control experiment was performed using the same flow rates in a micromodel with no precipi-

tates. By comparing the flowpath of fluorescein in micromodels with and without precipitates, permeability changes due to precipitation were identified.

Laser Confocal Imaging. Also after precipitation reached pseudo steady state, the micromodel pore network was entirely filled with the fluorescein solution and fluorescent confocal images were obtained using a Leica SP2 laser scanning confocal microscope at $0.21 \mu\text{m}$ in-plane resolution and $0.5 \mu\text{m}$ slice thickness to visualize the three-dimensional (3D) morphology of CaCO_3 precipitates.

Results and Discussions

Distribution, Morphology, and Mineralogy of CaCO_3 Crystal Precipitation. Images of micromodel pore spaces with calcium carbonate precipitates at pseudo-steady-state in each of the four experiments are shown in Figure 2. At all saturation states, crystal precipitation took place primarily in pore spaces along the center mixing zone between calcium and carbonate, with some precipitates forming around the edges of cylindrical posts. The transverse width of precipitates decreases with increasing saturation state. This zone is approximately two pore bodies wide at the two lower saturation states (Figure 2a,b), and less than one pore body at the two higher saturation states (Figure 2c,d). The total amount of CaCO_3 precipitation formed (based on the total number of pixels containing precipitate) decreases with increasing saturation state, with this amount approximately 5 times greater at the lowest saturation state than at the highest saturation state. For water-soluble species, higher saturation states are expected to result in more reactant mixing (greater concentration driving force), and more product formation. The opposite is true for CaCO_3 precipitation in the micromodels. This indicates that precipitates formed at higher saturation states are more densely packed (i.e., CaCO_3 precipitates have lower porosity) and limit transverse mixing. At pseudo-steady-state, the width of each precipitate zone in the transverse direction remained nearly constant along the flow direction in all experiments. This agrees with results from a continuum-scale study (21) in which mixing of calcium chloride and sodium carbonate in a 2D flow cell resulted in a calcium carbonate precipitate zone with uniform width.

Images of CaCO_3 precipitates in selected pore spaces at different saturation states, along with Raman spectra of selected CaCO_3 precipitates, are shown in Figure 3. At the lowest saturation state (i.e., $\Omega_c/\Omega_v = 3.4/2.8$ wrt calcite/vaterite, no aragonite detected), three morphologically distinct types of crystals precipitated along the center mixing zone as shown in Figure 2. These include (1) euhedral $50\text{--}100 \mu\text{m}$ diameter rhombohedra (Figure 3a inset) that exhibit distinct growth hillocks (41), (2) feather "shrub" aggregates

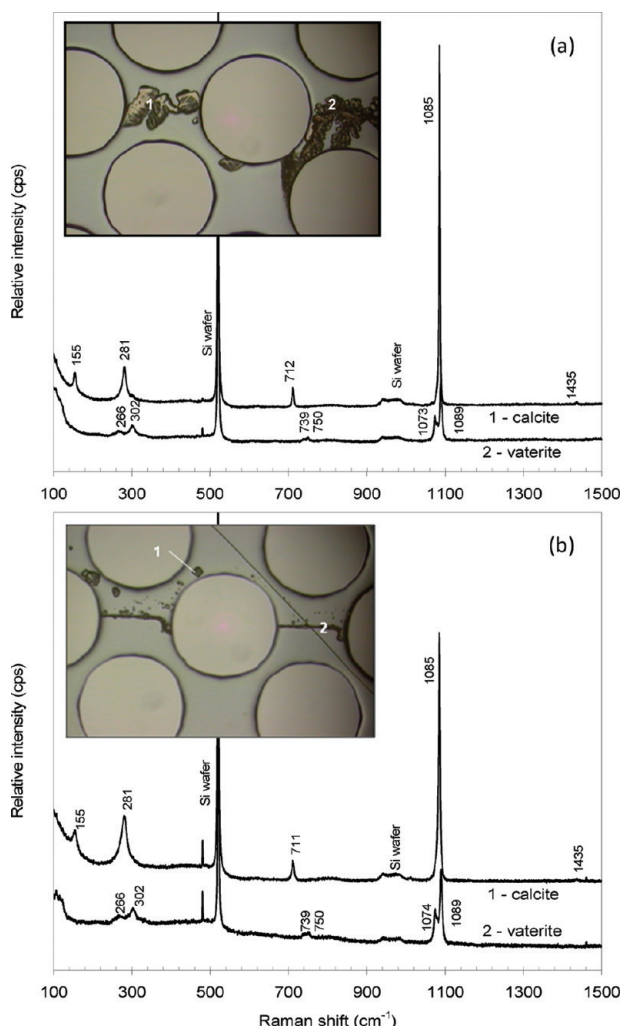


FIGURE 3. Raman back scattering spectra of CaCO_3 precipitates formed along the center mixing zone in the micromodel at (a) $\Omega_c/\Omega_v = 3.4/2.8$ and (b) $\Omega_c/\Omega_v = 5.2/4.5$.

(42, 43) composed of dendritically branched euhedral to subhedral columnar crystals, each of which reach 25 to 100 μm in length along the C-axis (Figure 3a inset), and (3) anhedral to spherical inclusion-rich microcrystalline crystals 5–20 μm in diameter (data not shown). Raman spectra of rhombohedra (Figure 3a) show medium to strong peaks at wavenumbers 155, 281, 712, and 1085 cm^{-1} that match those for calcite (20). Raman spectra of feather “shrub” aggregates (Figure 3a) and spherical microcrystalline crystals show medium to strong peaks at wavenumbers 266 and 302 (lattice/external mode), and 1073 and 1089 (internal mode) that match vaterite (20, 39, 40). Assuming crystals are uniform over the depth of the micromodel, precipitate shapes were used to estimate mineral forms, and vaterite and calcite accounted for approximately 80% and 20%, respectively, of the precipitates (by area).

At the highest saturation state (i.e., $\Omega_c/\Omega_v = 5.2/4.5$), the majority of precipitates were confined to thin dense zones distributed across pore bodies and around the perimeters of cylindrical posts along the center mixing zone, with isolated crystalline precipitates (both rhombohedra and spherical microcrystal) formed off center along the walls of cylindrical posts (Figure 3b inset). Raman spectra (Figure 3b) of solid precipitate lines and spherical microcrystals matched vaterite as characterized by peaks at wavenumbers 266, 302, 1074, and 1089 (20, 39, 40). Rhombohedra were found to be calcite. On the basis of the precipitate shapes, greater than 95% of the total precipitates were vaterite, with the remainder (<5%)

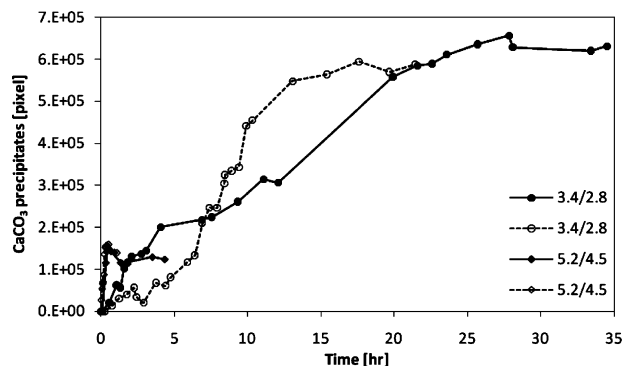


FIGURE 4. Overall CaCO_3 precipitation versus time in the micromodel at $\Omega_c/\Omega_v = 3.4/2.8$ and $\Omega_c/\Omega_v = 5.2/4.5$.

calcite. The amounts of vaterite and calcite for the intermediate saturation states were in between those in the low and high saturation state experiments.

Vaterite has the highest solubility among the three CaCO_3 polymorphs at ambient temperature and pressure, and is considered a metastable form that transforms to calcite (the more stable form) through Ostwald ripening. Carter et al. (20) showed that at 20 $^\circ\text{C}$, calcite accounts for <5% of the total precipitates in batch system after 2 min, and the percentage increased to 10–30% after 7 h. In the micromodel, Ca^{2+} and CO_3^{2-} are initially segregated and they only mix through diffusion. Their concentrations vary in space and Ω is at a maximum along the micromodel center zone, where CaCO_3 precipitates are preferentially formed. At high saturation state, we propose that rapid precipitation reduced mixing between Ca^{2+} and CO_3^{2-} and hence slowed phase transformation from vaterite to calcite.

CaCO_3 Precipitation Rate. The total number of pixels with CaCO_3 precipitate in the micromodel over time in the low and high saturation state experiments is shown in Figure 4 (results from intermediate saturation states are included in Figure S2 of the Supporting Information). From low to high saturation states, CaCO_3 precipitates initially formed at apparent rates of approximately 3.1×10^4 , 1.1×10^5 , 2.8×10^5 , and 4.0×10^5 pixels/h, and reached pseudo-steady-state after 25, 7, 1.5, and 0.5 h. Assuming uniform precipitate density in each pixel, and a bulk density of 2.6 g/mL (44), the precipitates formed at rates of approximately 6.8×10^{-7} , 2.5×10^{-6} , 6.0×10^{-6} , and 8.9×10^{-6} g/h, and the total amount of precipitates formed are 1.4×10^{-5} , 9.7×10^{-6} , 4.0×10^{-6} , and 2.7×10^{-6} g (from low to high saturation states). The increase in saturation state resulted in higher reactant concentrations in the mixing zone, and hence an increase in precipitation rate and a decrease in the amount of precipitates. It is important to note that precipitate density is likely not spatially uniform. However, the monotonic trends in precipitation rates and total precipitates formed at four different saturation states indicate changes in precipitate density would not change our general results.

The number of pixels containing CaCO_3 precipitate in three pores over time located upgradient, middle, and downgradient along the flow direction in the low and high saturation experiments are shown in Figure 5; results from the two intermediate saturation experiments are included in Figure S3 of the Supporting Information. At upgradient pores near the inlet, CaCO_3 precipitates were formed at higher rates (i.e., line slopes are greater) than the middle and downgradient pores; for example, at $\Omega_c/\Omega_v = 3.4/2.8$, the amount of precipitates in the upgradient pore space reached a maximum value after approximately 7.5 h, whereas in pores near the middle and downgradient sections, CaCO_3 precipitates reached a maximum after 20 h (Figure 5a). This behavior is general for the other two saturation states. The higher precipitation rate in the upgradient pores can be attributed

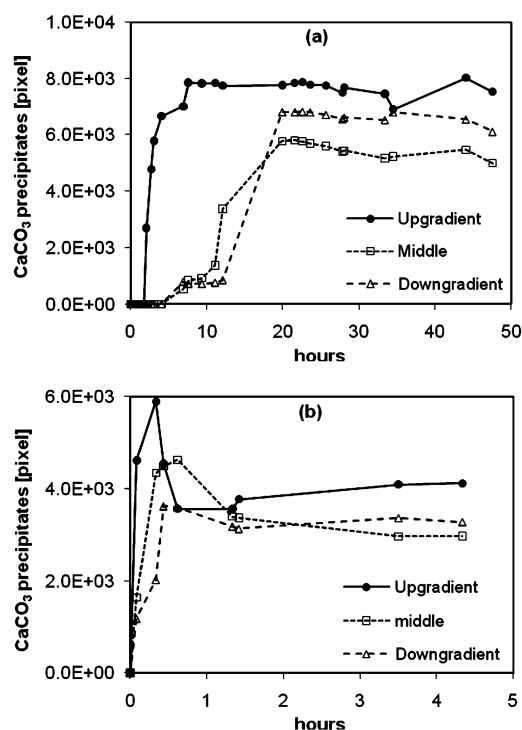


FIGURE 5. CaCO_3 precipitation versus time in selected pores at (a) $\Omega_c/\Omega_v = 3.4/2.8$ and (b) $\Omega_c/\Omega_v = 5.2/4.5$.

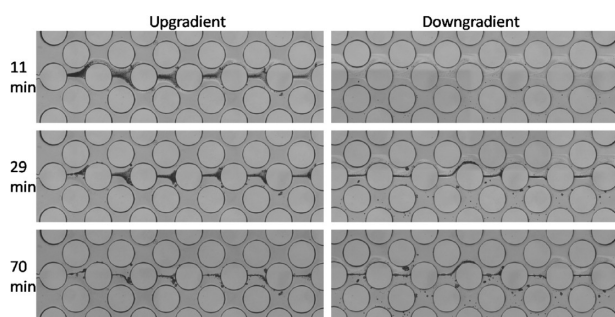


FIGURE 6. CaCO_3 precipitation and dissolution ($\Omega_c/\Omega_v = 5.2/4.5$).

to sharper reactant concentration gradients. Further, at high saturation state ($\Omega_c/\Omega_v = 5.2/4.5$), the amount of precipitates decreased after reaching a maximum and then leveled off (Figure 5b), which agrees with the overall trend in the entire micromodel.

Dissolution of CaCO_3 Precipitates. CaCO_3 precipitates initially formed near the inlet at high saturation states were found to dissolve over time; Figure 6 compares snapshots of selected pore spaces near the inlet and outlet at three different time points in the high saturation state experiment ($\Omega_c/\Omega_v = 5.2/4.5$). Near the inlet, the width of CaCO_3 precipitates decreased from 50 μm at 11 min to 10 μm at 70 min (Figure 6 left panels). We propose that this reduction occurs because the packing density of precipitates increases over time (i.e., porosity of CaCO_3 precipitates decreases with time), limiting mixing and reaction between solutes. In the pore space near the downgradient location, dissolution was also observed but to a lesser extent: from 23 μm at 29 min to 12 μm at 60 min (Figure 6 right panels).

Permeability Reduction. Changes in permeability as a result of CaCO_3 precipitation were not directly measured in the current study; instead, 3D laser confocal microscopy and fluorescent tracer tests were conducted to qualitatively evaluate permeability reduction. Laser confocal images in selected pores for two experiments ($\Omega_c/\Omega_v = 3.4/2.8$, 4.6/

3.9) show that CaCO_3 precipitates occupied the entire 20 μm depth of the micromodel, which indicates potential blockage of pore space. Crystals formed off-center or those formed along the cylinder walls also occupied the entire 20 μm depth in the micromodel, indicating crystal precipitation was uniform along the vertical dimension in the micromodel. Images of fluorescent tracer in the micromodels from a control experiment (no precipitates in the micromodel) and in the experiments after CaCO_3 precipitation reached pseudo-steady-state were obtained using a 2:1 flow rate of tracer: water. The tracer solution occupied approximately 2/3 of the transverse cross-sectional area of the micromodel in the absence of CaCO_3 precipitates, but only half of the transverse cross-sectional area of the micromodels after CaCO_3 precipitates formed at each of the four saturation states. This confirms blockage of the pore space by CaCO_3 precipitates. Images that support these results are in the Supporting Information (Figures S4 and S5).

Implications. Results from this study indicate mineral precipitation reactions in subsurface sedimentary systems can limit transverse mixing between reactive fluids. Different polymorphs of CaCO_3 crystal precipitation can form along transverse mixing zones depending on the solution saturation state, and these result in a decrease in porosity, a reduction in mixing between reactive species which limits mineral phase transformation, and a reduction in effective permeability. Implications of this study on processes occurring in subsurface sedimentary systems include the following: (i) along the fringes of CO_2 plumes created during geological carbon sequestration mineral precipitation may decrease the effective permeability and result in a buildup of pressure and potential loss of injectivity, (ii) precipitates formed along plume margins during in situ remediation can decrease reactant mixing and decrease cleanup efficiency, (iii) mixing and precipitation of leachate from nuclear waste repository with surrounding groundwater can be an effective way of storing radioactive wastes (26). Experiments performed in this work represent a simplified scenario, more complex solution conditions (i.e., presence of competing cations, different ionic strength, pH, temperature, and pressure) and physical and chemical heterogeneity in 3D porous media are expected. Further study is needed to evaluate how these complexities affect mineral precipitation, and corresponding changes in reactant mixing, mineralogy, porosity, and permeability.

Acknowledgments

This work was supported by the Office of Science (BER), U.S. Department of Energy (DOE), Grant No. DE-FG02-06ER64207, by a Graduate Assistance in Areas of National Need fellowship from the U.S. Department of Education for Karl Dehoff, Grant P200A060190, by the Pacific Northwest National Laboratory Directed Research and Development Program under PNNL's Carbon Sequestration Initiative, and by LDRD 82552-001-10 sponsored by the Los Alamos National laboratory. A portion of the experiments were conducted in the William R. Wiley Environmental Molecular Sciences Laboratory, a United States Department of Energy (DOE) scientific user facility operated for the DOE by PNNL. We thank Jon Ekman at the Beckman Imaging Technology Group, University of Illinois, for help with laser confocal imaging.

Supporting Information Available

High resolution image (0.65 μm) used for mineral analysis (Figure S1), total and region specific precipitation versus time from micromodel experiments at intermediate saturation states (Figure S2, S3), laser confocal microscope images of precipitates in selected pores (Figure S4), and images of a fluorescent tracer before and after CaCO_3 precipitation in

the micromodel (Figure S5). This material is available free of charge via the Internet at <http://pubs.acs.org>.

Literature Cited

- Audigane, P.; Gaus, I.; Czernichowski-Lauriol, I.; Pruess, K.; Xu, T. F. Two-dimensional reactive transport modeling of CO₂ injection in a saline Aquifer at the Sleipner site, North Sea. *Am. J. Sci.* **2007**, *307* (7), 974–1008.
- Doughty, C.; Freifeld, B. M.; Trautz, R. C. Site characterization for CO₂ geologic storage and vice versa: The Frio brine pilot, Texas, USA as a case study. *Environ. Geol.* **2008**, *54* (8), 1635–1656.
- Nicot, J.-P. Evaluation of large-scale CO₂ storage on fresh-water sections of aquifers: An example from the Texas Gulf Coast Basin. *Int. J. Greenhouse Gas Control* **2008**, *2* (4), 582–593.
- Doughty, C. Investigation of CO₂ plume behavior for a large-scale pilot test of geologic carbon storage in a saline formation. *Transp. Porous Media* **2010**, *82* (1), 49–76.
- Person, M.; Banerjee, A.; Rupp, J.; Medina, C.; Lichtner, P.; Gable, C.; Pawar, R.; Celia, M.; McIntosh, J.; Bense, V. Assessment of basin-scale hydrologic impacts of CO₂ sequestration, Illinois basin. *Int. J. Greenhouse Gas Control* **2010**, *4* (5), 840–854.
- IPCC Special Report on Carbon Dioxide Capture and Storage, Cambridge University Press: Cambridge, U.K., 2005; 442 pp.
- CCS Guidelines: Guidelines for Carbon Dioxide Capture, Transport and Storage, W. R. I.: Washington, DC, 2008; 144 p.
- National Energy Technology Laboratory. *Carbon Sequestration Atlas of the United States and Canada*; U.S. Department of Energy, Office of Fossil Energy: Washington DC, 2007; 90 p.
- Rosenbauer, R. J.; Koksalan, T.; Palandri, J. L. Experimental investigation of CO₂-brine-rock interactions at elevated temperature and pressure: Implications for CO₂ sequestration in deep-saline aquifers. *Fuel Process. Technol.* **2005**, *86* (14–15), 1581–1597.
- Kharaka, Y. K.; Thordsen, J. J.; Hovorka, S. D.; Nance, H. S.; Cole, D. R.; Phelps, T. J.; Knauss, K. G. Potential environmental issues of CO₂ storage in deep saline aquifers: Geochemical results from the Frio-I Brine Pilot test, Texas, USA. *Appl. Geochem.* **2009**, *24* (6), 1106–1112.
- Johnson, J. W.; Nitao, J. J.; Knauss, K. G. Reactive transport modelling of CO₂ storage in saline aquifers to elucidate fundamental processes, trapping mechanisms and sequestration partitioning. In *Geological Storage of Carbon Dioxide*; Baines, S. J., Worden, R. H., Eds.; Geological Society, London, Special Publications: 2004; Vol. 233.
- Ennis-King, J.; Paterson, L. Role of convective mixing in the long-term storage of carbon dioxide in deep saline formations. *SPE J.* **2005**, *10* (3), 349–356.
- Ennis-King, J.; Paterson, L. Coupling of geochemical reactions and convective mixing in the long-term geological storage of carbon dioxide. *Int. J. Greenhouse Gas Control* **2007**, *1* (1), 86–93.
- Giammar, D. E.; Bruant, R. G.; Peters, C. A. Forsterite dissolution and magnesite precipitation at conditions relevant for deep saline aquifer storage and sequestration of carbon dioxide. *Chem. Geol.* **2005**, *217* (3–4), 257–276.
- Luquot, L.; Gouze, P. Experimental determination of porosity and permeability changes induced by injection of CO₂ into carbonate rocks. *Chem. Geol.* **2009**, *265* (1–2), 148–159.
- Liteanu, E.; Spiers, C. J. Influence of pore fluid salt content on compaction creep of calcite aggregates in the presence of supercritical CO₂. *Chem. Geol.* **2009**, *265* (1–2), 134–147.
- Xu, T. F.; Apps, J. A.; Pruess, K. Reactive geochemical transport simulation to study mineral trapping for CO₂ disposal in deep arenaceous formations. *J. Geophys. Res., [Solid Earth]* **2003**, *108* (B2), DOI: 10.1029/2002JB001979.
- Xu, T. F.; Apps, J. A.; Pruess, K. Numerical simulation of CO₂ disposal by mineral trapping in deep aquifers. *Appl. Geochem.* **2004**, *19* (6), 917–936.
- Matter, J. M.; Kelemen, P. B. Permanent storage of carbon dioxide in geological reservoirs by mineral carbonation. *Nat. Geosci.* **2009**, *2* (12), 837–841.
- Carteret, C.; Dandeu, A.; Moussaoui, S.; Muhr, H.; Humbert, B.; Plasari, E. Polymorphism studied by lattice phonon raman spectroscopy and statistical mixture analysis method. Application to calcium carbonate polymorphs during batch crystallization. *Cryst. Growth Des.* **2009**, *9* (2), 807–812.
- Tartakovsky, A. M.; Redden, G.; Lichtner, P. C.; Scheibe, T. D.; Meakin, P. Mixing-induced precipitation: Experimental study and multiscale numerical analysis. *Water Resour. Res.* **2008**, *44* (6), DOI: 10.1029/2006WR005725.
- Katz, G. E.; Berkowitz, B.; Guadagnini, A.; Saaltink, M. W., Experimental and modeling investigation of multicomponent reactive transport in porous media. *J. Contam. Hydrol.* **2010**, DOI: 10.1016/j.jconhyd.2009.11.002.
- Wigand, M.; Kaszuba, J. P.; Carey, J. W.; Hollis, W. K. Geochemical effects of CO₂ sequestration on fractured wellbore cement at the cement/caprock interface. *Chem. Geol.* **2009**, *265* (1–2), 122–133.
- Yabusaki, S.; Cantrell, K.; Sass, B.; Steefel, C. Multicomponent reactive transport in an in situ zero-valent iron cell. *Environ. Sci. Technol.* **2001**, *35* (7), 1493–1503.
- Keith, D. C.; Harrison, W. J.; Wendlandt, R. F.; Daniels, E. J. Mineralogical responses of siliciclastic carbonate-cemented reservoirs to steamflood enhanced oil recovery. *Appl. Geochem.* **1998**, *13* (4), 491–507.
- Nicot, J. P. Methodology for bounding calculations of nuclear criticality of fissile material accumulations external to a waste container at Yucca Mountain, Nevada. *Appl. Geochem.* **2008**, *23* (8), 2065–2081.
- Gramling, C. M.; Harvey, C. F.; Meigs, L. C. Reactive transport in porous media: A comparison of model prediction with laboratory visualization. *Environ. Sci. Technol.* **2002**, *36* (11), 2508–2514.
- Oates, P. M.; Harvey, C. F. A colorimetric reaction to quantify fluid mixing. *Exp. Fluids* **2006**, *41* (5), 673–683.
- Willingham, T. W.; Werth, C. J.; Valocchi, A. J. Evaluation of the effects of porous media structure on mixing-controlled reactions using pore-scale modeling and micromodel experiments. *Environ. Sci. Technol.* **2008**, *42* (9), 3185–3193.
- Willingham, T.; Zhang, C.; Werth, C. J.; Valocchi, A. J.; Oostrom, M.; Wietsma, T. W. Using dispersivity values to quantify the effects of pore-scale flow focusing on enhanced reaction along a transverse mixing zone. *Adv. Water Resour.* **2010**, *33* (4), 525–535.
- Steefel, C. I.; Lichtner, P. C. Diffusion and reaction in rock matrix bordering a hyperalkaline fluid-filled fracture. *Geochim. Cosmochim. Acta* **1994**, *58* (17), 3595–3612.
- Kang, Q. J.; Zhang, D. X.; Chen, S. Y. Simulation of dissolution and precipitation in porous media. *J. Geophys. Res., [Solid Earth]* **2003**, *108* (B10), DOI: 10.1029/2003JB002504.
- Emmanuel, S.; Berkowitz, B. Mixing-induced precipitation and porosity evolution in porous media. *Adv. Water Resour.* **2005**, *28* (4), 337–344.
- Li, L.; Peters, C. A.; Celia, M. A. Effects of mineral spatial distribution on reaction rates in porous media. *Water Resour. Res.* **2007**, *43* (1), DOI: 10.1029/2005WR004848.
- Zhang, C. Y.; Kang, Q. J.; Wang, X.; Zilles, J. L.; Muller, R. H.; Werth, C. J. Effects of Pore-Scale Heterogeneity and Transverse Mixing on Bacterial Growth in Porous Media. *Environ. Sci. Technol.* **2010**, *44* (8), 3085–3092.
- Stumm, W. *Chemistry of the Solid-Water Interface*; John Wiley & Sons, Inc: New York, 1992.
- Kandianis, M. T.; Fouke, B. W.; Johnson, R. W.; Veysey, J.; Inskeep, W. P. Microbial biomass: A catalyst for CaCO₃ precipitation in advection-dominated transport regimes. *Geol. Soc. Am. Bull.* **2008**, *120* (3–4), 442–450.
- Chomsurin, C.; Werth, C. J. Analysis of pore-scale nonaqueous phase liquid dissolution in etched silicon pore networks. *Water Resour. Res.* **2003**, *39* (9), DOI: 10.1029/2002WR001643.
- Behrens, G.; Kuhn, L. T.; Ueb, R.; Heuer, A. H. Raman-spectra of vaterite calcium-carbonate. *Spectrosc. Lett.* **1995**, *28* (6), 983–995.
- Veinott, G. I.; Porter, T. R.; Nasdala, L. Using Mg as a proxy for crystal structure and Sr as an indicator of marine growth in vaterite and aragonite otoliths of aquaculture rainbow trout. *Trans. Am. Fish. Soc.* **2009**, *138* (5), 1157–1165.
- Paquette, J.; Reeder, R. J. New type of compositional zoning in calcite—Insights into crystal-growth mechanisms. *Geology* **1990**, *18* (12), 1244–1247.
- Dickson, J. A. D. Crystal-growth diagrams as an aid to interpreting the fabrics of calcite aggregates. *J. Sediment. Petrol.* **1993**, *63* (1), 1–17.
- Jones, B.; Renaut, R. W.; Owen, R. B.; Torfason, H. Growth patterns and implications of complex dendrites in calcite travertines from Lysuholl, Snæfellsnes, Iceland. *Sedimentology* **2005**, *52* (6), 1277–1301.
- Wollenweber, J.; Alles, S.; Busch, A.; Krooss, B. M.; Stanjek, H.; Litke, R. Experimental investigation of the CO₂ sealing efficiency of caprocks. *Int. J. Greenh. Gas Con.* **2010**, *4* (2), 231–241.

ES1019788

SUPPORTING INFORMATION

**Pore-Scale Study of Transverse Mixing Induced CaCO₃
Precipitation and Permeability Reduction in a Model Subsurface
Sedimentary System**

Changyong Zhang¹, Karl Dehoff², Nancy Hess¹, Mart Oostrom³, Thomas W. Wietsma⁴, Albert J. Valocchi², Bruce W. Fouke⁵, Charles J. Werth^{*2}

¹Fundamental and Computational Sciences Directorate, Pacific Northwest National Laboratory, Richland WA 99352

²Department of Civil and Environmental Engineering, University of Illinois at Urbana-Champaign, 205 N Mathews Ave, Urbana IL 61801

³Energy and Environment Directorate, Pacific Northwest National Laboratory, Richland, WA 99354

⁴Environmental Molecular Sciences Laboratory, Pacific Northwest National Laboratory, Richland, WA 99354

⁵Department of Geology, University of Illinois at Urbana-Champaign, 1301 W Green St, Urbana IL 61801

* Corresponding author e-mail: werth@illinois.edu; phone: 217-333-3822; fax: 217-333-6968.

Journal: Environmental Science & Technology

Number of pages: 6

Number of Figures: 5

Date prepared: August 12, 2010

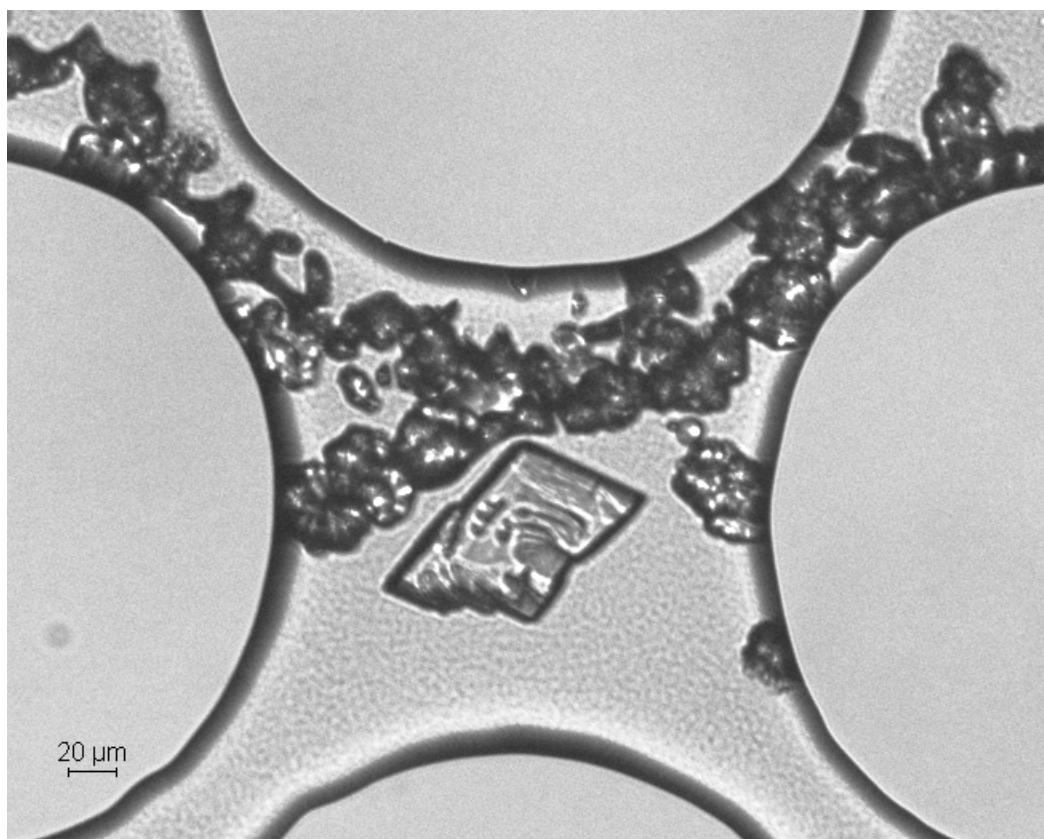


Figure S1. High resolution image (0.65 μm) used for mineral form analysis.

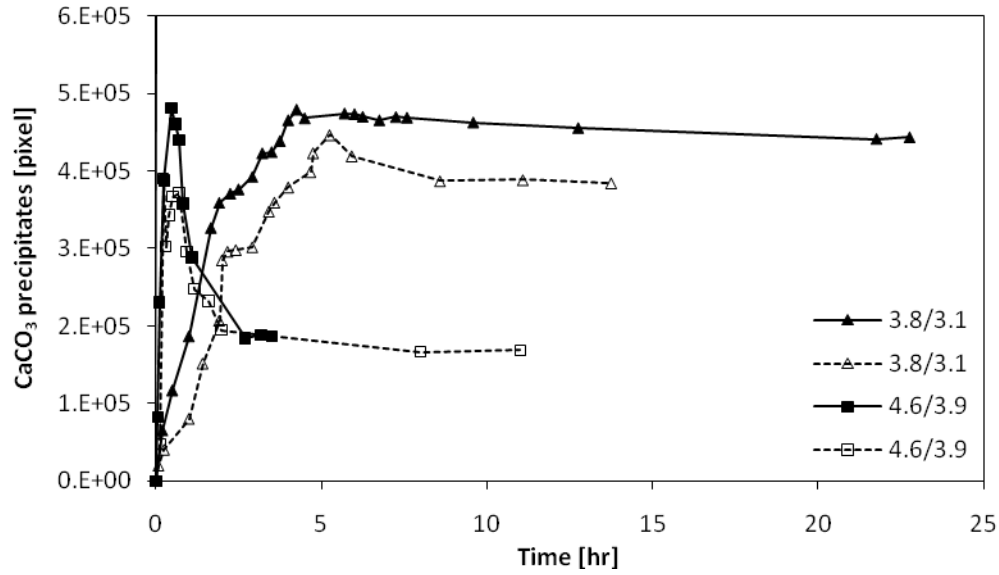


Figure S2. Overall precipitation rate in the micromodel at $\Omega_c/\Omega_v = 3.8/3.1$, and $\Omega_c/\Omega_v = 4.6/3.9$.

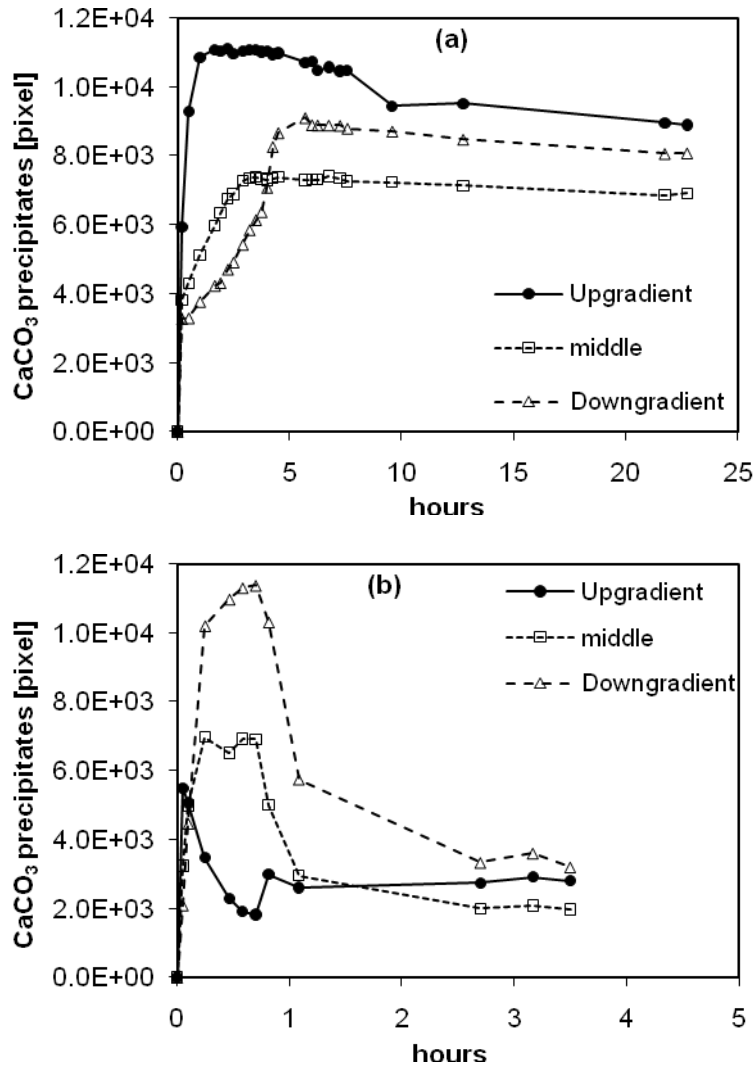


Figure S3. CaCO₃ precipitation rate at selected pore space at (a) $\Omega_c/\Omega_v = 3.8/3.1$, and (b), $\Omega_c/\Omega_v = 4.6/3.9$.

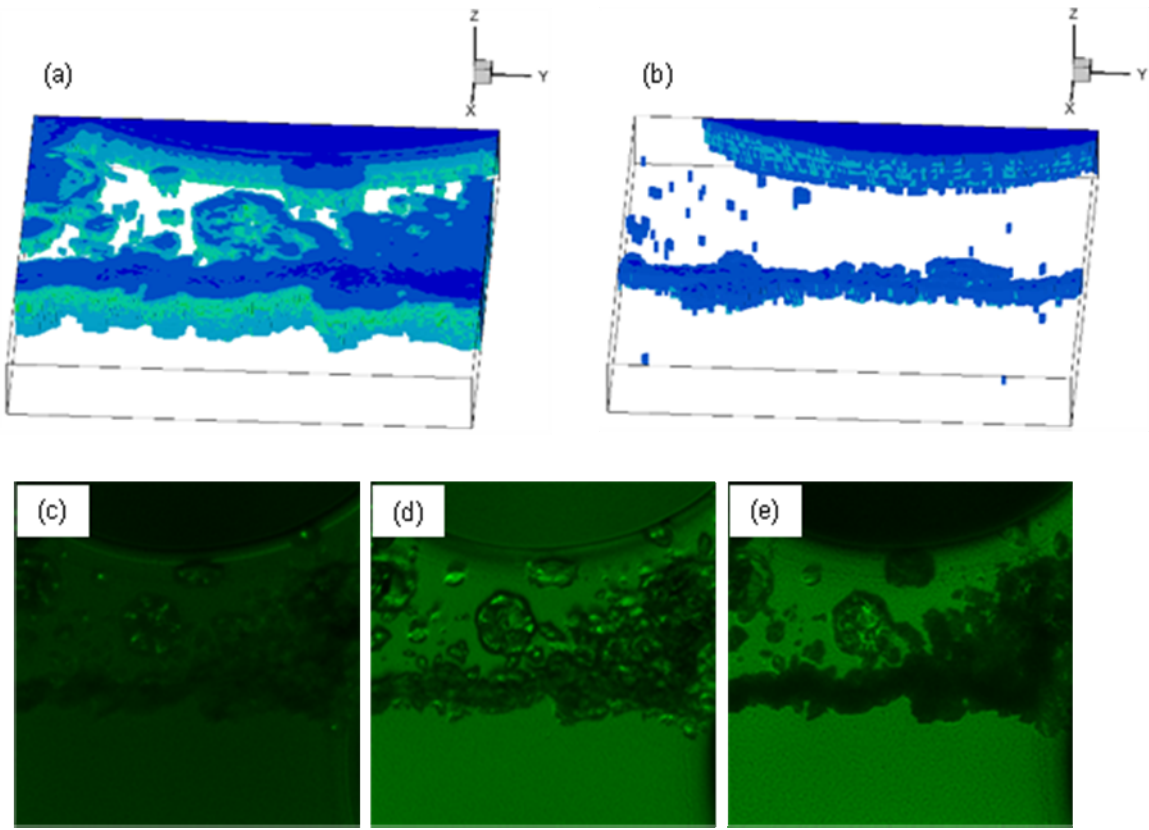


Figure S4. 3D confocal images of CaCO₃ precipitates in a micromodel pore at (a) $\Omega_c/\Omega_v = 3.4/2.8$ and (b) $\Omega_c/\Omega_v = 4.6/3.9$, and (c-e) 2D confocal images at selected depths at $\Omega_c/\Omega_v = 3.4/2.8$.

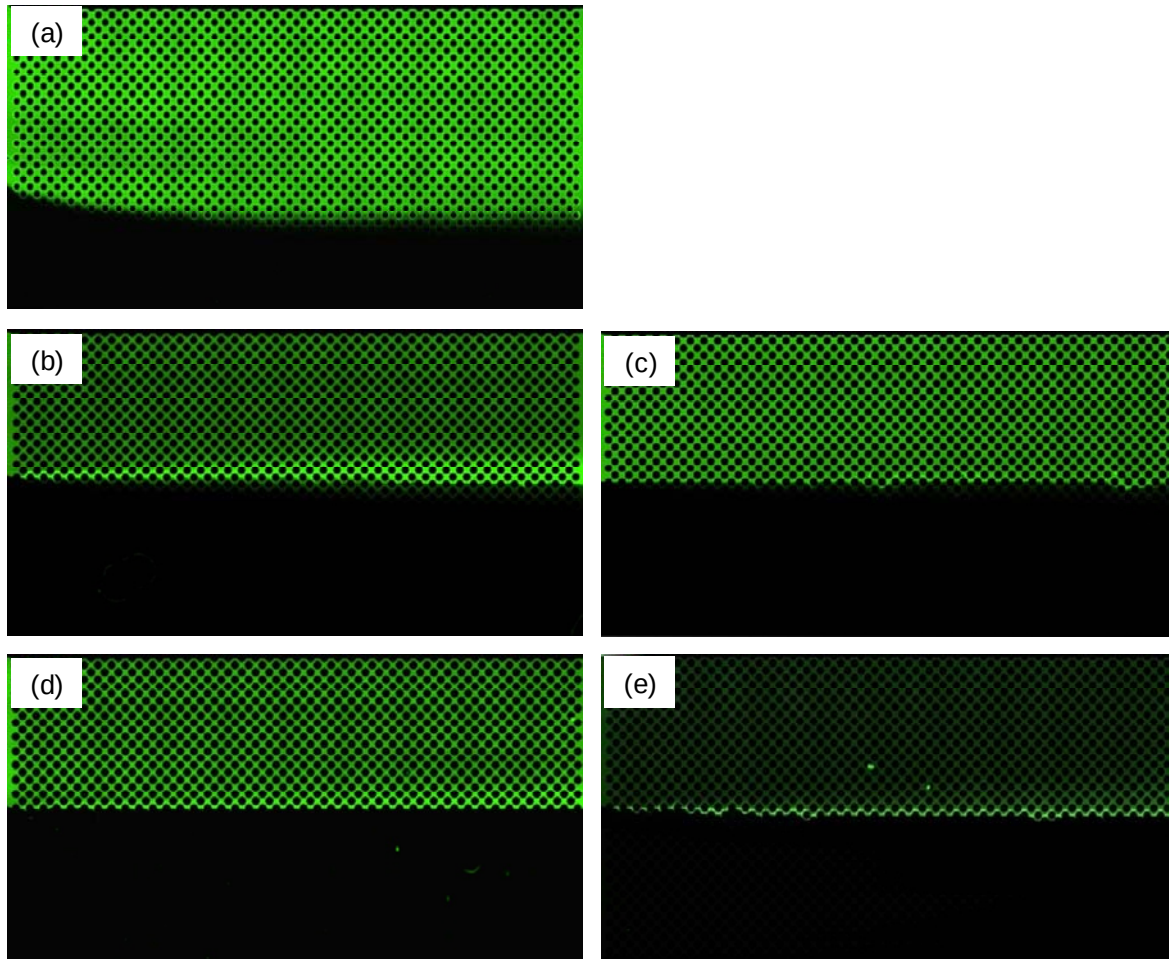


Figure S5. Images of the fluorescent tracer in the micromodel before CaCO_3 precipitates formed (a), and after CaCO_3 precipitates formed at $\Omega_c/\Omega_v = 3.4/2.8$ (b), $\Omega_c/\Omega_v = 3.8/3.1$ (c), $\Omega_c/\Omega_v = 4.6/3.9$ (d), and $\Omega_c/\Omega_v = 5.2/4.5$ (e).

Pore-scale simulation of mixing-induced calcium carbonate precipitation and dissolution in a microfluidic pore network

Hongkyu Yoon,¹ Albert J. Valocchi,² Charles J. Werth,² and Thomas Dewers¹

Received 23 July 2011; revised 12 December 2011; accepted 13 January 2012; published 18 February 2012.

[1] We develop a 2-D pore scale model of coupled fluid flow, reactive transport, and calcium carbonate (CaCO_3) precipitation and dissolution. The model is used to simulate transient experimental results of CaCO_3 precipitation and dissolution under supersaturated conditions in a microfluidic pore network (i.e., micromodel) in order to improve understanding of coupled reactive transport systems perturbed by geological CO_2 injection. In the micromodel, precipitation is induced by transverse mixing along the centerline in pore bodies. The reactive transport model includes the impact of pH upon carbonate speciation and a CaCO_3 reaction rate constant, the effect of changing reactive surface area upon the reaction, and the impact of pore blockage from CaCO_3 precipitation on diffusion and flow. Overall, the pore scale model qualitatively captured the precipitate morphology, precipitation rate, and maximum precipitation area using parameter values from the literature. In particular, we found that proper estimation of the effective diffusion coefficient (D_{eff}) and the reactive surface area is necessary to adequately simulate precipitation and dissolution rates. In order to match the initial phase of fast precipitation, it was necessary to consider the top and bottom of the micromodel as additional reactive surfaces. In order to match a later phase when dissolution occurred, it was necessary to increase the dissolution rate compared to the precipitation rate, but the simulated precipitate area was still higher than the experimental results after ~ 30 min, highlighting the need for future study. The model presented here allows us to simulate and mechanistically evaluate precipitation and dissolution of CaCO_3 observed in a micromodel pore network. This study leads to improved understanding of the fundamental physicochemical processes of CaCO_3 precipitation and dissolution under far-from-equilibrium conditions.

Citation: Yoon, H., A. J. Valocchi, C. J. Werth, and T. Dewers (2012), Pore-scale simulation of mixing-induced calcium carbonate precipitation and dissolution in a microfluidic pore network, *Water Resour. Res.*, 48, W02524, doi:10.1029/2011WR011192.

1. Introduction

[2] The formation of calcium carbonate plays an important role in many natural and engineered systems, including geological CO_2 storage [Johnson *et al.*, 2004; Rochelle *et al.*, 2004], biogeochemical behavior in marine environments [Riebesell *et al.*, 2000], biomineralization of products from microorganisms [De Yoreo and Vekilov, 2003], growth of calcite concretions in fractured and porous media [Mozley and Davis, 2005], and the formation of scale in boilers and cooling towers [Richter *et al.*, 1989]. Knowledge of the kinetics and mechanisms of crystal growth facilitates prediction of mineralization associated with these processes, and can guide the design of engineered systems. Within hydrogeochemical, engineered, or biological systems, mineralization processes occur at different spatial and temporal scales. In particular, geochemical

reactions in porous media occur at the molecular scale, and mineral products affect flow and mixing processes at the pore and continuum scale. These have consequences for predicting reactive transport processes in porous media, such as those during geological storage of CO_2 . Mineral precipitation and dissolution has been extensively studied in well-mixed batch systems; however, there have been few investigations for transport conditions where mass transfer and dispersion limits the mixing of reactants. As a result, a mechanistic understanding of mineral formation and dissolution rates in porous media is lacking, thus limiting the accuracy of quantitative assessments required for scale-up to field systems.

[3] There is extensive literature that details indirect measurement approaches (e.g., measuring solution chemistry, weight, and/or size) to quantify calcium carbonate (CaCO_3) crystallization in well-mixed systems under a wide range of chemical compositions and conditions [Plummer *et al.*, 1978; Inskeep and Bloom, 1985; Blum and Lasaga, 1987; Hostomsky and Jones, 1991; Shiraki and Brantley, 1995]. These studies have been used to develop quantitative representations of calcite growth rate laws commonly used in reactive transport models for porous and fractured media. However, these models have not been confirmed through

¹Geomechanics Department, Sandia National Laboratories, Albuquerque, New Mexico, USA.

²Civil and Environmental Engineering, University of Illinois at Urbana-Champaign, Urbana, Illinois, USA.

observations of the actual growth processes at mineral surfaces. Recently, in situ measurement techniques have been used to elucidate crystal growth mechanisms from highly supersaturated solutions in well-mixed systems [Pontoni *et al.*, 2003; Rieger *et al.*, 2007; Liu *et al.*, 2010; Kellermeier *et al.*, 2010]. In particular, these works investigated the role of nano-size amorphous calcium carbonate (ACC) in the growth of calcium carbonate crystals. ACC is the thermodynamically least stable among all calcium carbonate (CaCO_3) polymorphs, followed by vaterite, aragonite, and calcite. Under highly supersaturated conditions, ACC nanoparticles with a size of ~ 30 nm are formed within a minute; subsequently, these particles grow and transform with time to more stable crystalline forms such as vaterite crystals [Pouget *et al.*, 2009; Liu *et al.*, 2010], and eventually calcite [Ogino *et al.*, 1987; Rieger *et al.*, 2007].

[4] Significant efforts have also been made to elucidate crystal growth mechanisms from supersaturated solutions under representative hydrodynamic conditions using in situ measurement techniques. Mineral growth and dissolution were measured using in situ atomic force microscopy (AFM) [Teng *et al.*, 2000; De Yereo *et al.*, 2009; Stack and Grantham, 2010] and X-ray computed microtomography [Gouze *et al.*, 2003; Noiriel *et al.*, 2004], and mineral morphology and type were measured using a combination of reflected differential interference contrast (DIC) microscopy and backscattering Raman-spectroscopy [Zhang *et al.*, 2010]. Stack and Grantham [2010] determined using AFM that growth rates of calcite and calcite morphology are strongly affected by the ratio of calcium to carbonate concentration, implying that calcite growth patterns will be affected in a spatially varying concentration environment such as porous media. These findings are supported by results from Zhang *et al.* [2010], where Raman spectroscopy was used to show that more vaterite relative to calcite was formed at higher calcium and carbonate concentrations in a microfluidic pore network, and that vaterite crystals were smaller than calcite crystals. These novel experimental and theoretical results under hydrodynamic conditions have the potential to enhance our ability to simulate reactive transport in the subsurface.

[5] A number of pore scale models for flow and reactive transport relevant to geological storage of CO_2 have been developed using lattice Boltzmann (LB) [Kang *et al.*, 2003, 2006, 2010], smooth particle hydrodynamics (SPH) [Tartakovsky *et al.*, 2007, 2008, 2009], direct numerical simulation [Flukiger and Bernard, 2009], and pore network models [Li *et al.*, 2006; Kim *et al.*, 2011]. Both LB and SPH are suitable for complex geometries and a variety of reactive surface boundaries, and have been applied to fluid flow and multicomponent reactive transport problems. These modeling studies characterized qualitative features of reactive transport in a simple numerical domain over a wide range of dimensionless Peclet (Pe) and Damköhler (Da) numbers [Kang *et al.*, 2006, 2007; Tartakovsky *et al.*, 2007, 2008]. In particular, the SPH model was used to characterize mixing-induced precipitation in a meter-long tank experiment over a range of the Peclet and Damköhler numbers where the characteristic length scale of precipitation was on the order of the average sand grain diameter along the mixing line [Tartakovsky *et al.*, 2008]. However, the SPH model did not match changes in pH, carbonate species

concentrations, or the saturation index value in the tank experiment. Pore network reactive flow models [Li *et al.*, 2006; Kim *et al.*, 2011] remain a practical approach for simulation since they are flexible, computationally efficient, and can incorporate reaction rate laws and crystal growth patterns into the modeling framework. However, these attempts are based on simplified assumptions for crystal growth at the fluid-reaction interface, and do not fully account for pore scale processes under hydrodynamic conditions occurring in porous media.

[6] Recent developments of in situ measurement techniques for pore scale-reactive transport experiments provide a unique opportunity to test and validate pore scale-modeling approaches. For example, recent attempts to simulate CaCO_3 precipitation and dissolution in porous media [Emmanuel and Berkowitz, 2005; Tartakovsky *et al.*, 2008; Kang *et al.*, 2010] revealed the importance of porous media structure and a heterogeneous reaction rate constant. In porous and fractured media, the numerical studies suggest that diffusion-controlled precipitate can armor fractures, and limit the amount of mixing and subsequent reaction [Steefel and Lichtner, 1994, 1998; Kang *et al.*, 2010]. However, the theory that captures these effects has not been quantitatively compared against experimental data. Also, several key processes have not been evaluated by pore scale models, including the effect of mineral formation on diffusion [Navarre-Sitchler *et al.*, 2009] and reactive surface area [Navarre-Sitchler and Brantley, 2007]. Pore scale or micro-scale models are typically used to gain fundamental insights into physical behavior so that more accurate and robust upscaled constitutive models can be developed for the continuum scale [Whitaker, 1999; Wood, 2009; Kim *et al.*, 2011]. The development of pore scale models also provides the opportunity to more accurately simulate results at practical length scales in a hybrid approach in which pore- and continuum-scale equations are combined using either a domain-decomposition approach [Balhoff *et al.*, 2008; Battisto *et al.*, 2011] or dimension-reduction methods based on numerical closure with an explicit averaging of information from pore scale simulations [Tartakovsky and Scheibe, 2011].

[7] The objectives of this study are to present a new pore scale-reactive transport model for calcium carbonate precipitation and dissolution, to evaluate it with published experimental results from a microfluidic pore network experiment, and to identify mechanisms that are poorly captured with existing theory. The new pore scale model consists of three coupled modules: a lattice Boltzmann (LB) module for Stokes flow in the pore network, a finite volume multicomponent reactive transport module for the advection and diffusion of dissolved species, and a new pore scale crystal growth module that determines the amount of precipitate in the pore space. Model results are compared to experimental results by Zhang *et al.* [2010], who measured calcium carbonate precipitation and dissolution in a two-dimensional pore network (i.e., micromodel) along a transverse mixing zone between influent solutions of calcium chloride and sodium carbonate. We evaluate how the effective diffusion coefficient, reaction rate, and reaction surface area affect agreement between model simulations and experimental results. The implications of the pore scale modeling results are discussed.

2. Micromodel Experiment

[8] Homogeneous micromodels with a uniform distribution of cylinders were used to perform transverse-mixing induced CaCO_3 precipitation and dissolution experiments [Zhang *et al.*, 2010; Dehoff, 2010]. A detailed description of the fabrication and assembly process is given by Chomsurin and Werth [2003] and Willingham *et al.* [2008], and details of experimental results are given by Zhang *et al.* [2010]. The micromodel pore network is $1 \text{ cm} \times 2 \text{ cm}$ (width $\times$ length), consisting of cylindrical posts $300 \mu\text{m}$ in diameter, $180 \mu\text{m}$ pore bodies, and $40 \mu\text{m}$ pore throats. The depth of a micromodel pore is $20 \mu\text{m}$. Porosity is ~ 0.39 and water was continuously injected through two separate inlets, resulting in a constant Darcy velocity of 1.25 cm min^{-1} .

[9] The experiments consisted of injecting solutions containing equimolar concentrations of CaCl_2 and Na_2CO_3 through two separate inlets, transverse diffusion caused the two solutions to mix along the center of the micromodel, resulting in CaCO_3 precipitation. In this work, we focus on modeling CaCO_3 precipitation and dissolution in the first three pore bodies closest to the inlets of the pore network. We evaluated transient CaCO_3 precipitation and dissolution with 25 mM influent concentrations of total calcium and carbonate until the amount of precipitate reached quasi-steady state (i.e., the precipitate area was constant over time).

[10] An experimental image of precipitates at steady state for the 25 mM inlet concentrations is shown in Figure 1a. Precipitation occurred mostly along the centerline within one pore space transverse to the primary flow direction, with some large crystals off the centerline. The transverse width of the precipitate line tends to an increase with distance from the inlet. Two observations relevant to this work are that precipitation occurred upstream prior to downstream, and the precipitate is slightly asymmetric in the transverse direction downstream as shown in the first

pore body. Using backscattering Raman spectroscopy, Zhang *et al.* [2010] identified that larger crystals are calcite, and clusters of smaller particles along the centerline are vaterite. In this study, only vaterite formation is considered since it is the dominant precipitate phase under the experimental conditions considered.

[11] The experimental images used in this study were taken using reflected differential interference contrast microscopy [Zhang *et al.*, 2010]. The image intensity of CaCO_3 precipitates (shown in black or dark gray) is lower than that of the cylinder and pore spaces, but a shadow around the cylinder has a similar intensity (Figure 1a). In this study, new masks were created to cover the cylindrical posts and their shadows over the image region considered (i.e., the first three pore spaces), and then a threshold value was determined for each image to calculate the precipitate areas (i.e., the number of pixels). This method results in a 2-D precipitate area. The measured precipitate areas in the first pore body and entire numerical domain (i.e., the first three pore spaces) are shown in Figure 1b. An important result for this study is that the precipitate area increases rapidly due to fast precipitation, and then decreases due to fast dissolution within 13 min until a relatively constant but lower plateau is reached over 3–4 h. We denote the rapid increase in precipitate area followed by a decrease *overshoot*. In this study, the characteristic of the precipitate area over time (i.e., overshoot) and patterns of the precipitate distribution in the 2-D micromodel are investigated.

[12] A basic assumption of this work is that many of the important features in the micromodel experiments can be captured with a 2-D numerical model, because all experimental images were measured in 2-D, and laser confocal microscopy images showed that mineral precipitates span the entire depth of micromodel pores at the end of the experiment [Zhang *et al.*, 2010]. In addition, because of the

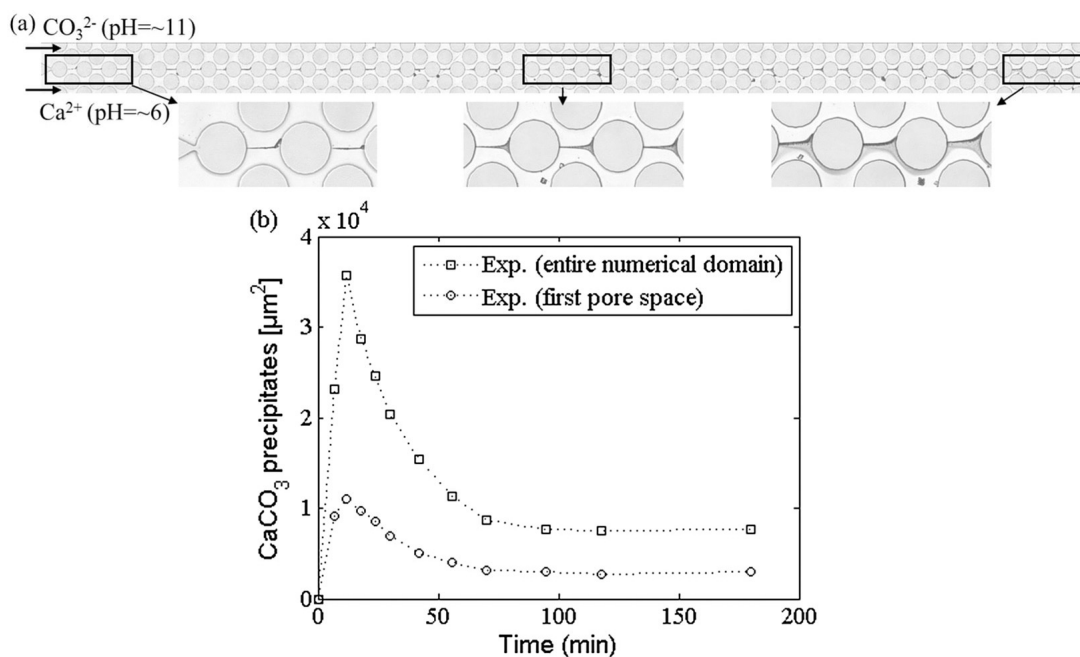


Figure 1. (a) Image of CaCO_3 precipitates in the center of the micromodel at 118 min for 25 mM influent concentrations and (b) CaCO_3 precipitate area over time.

shallow depth (20 μm) of the micromodel, the diffusion time-scale for solute concentration to be uniform with depth is relatively small (~ 0.1 s) compared to the fluid residence time (~ 2 min). Hence, the reaction is dominantly affected by horizontal transverse-mixing of two reactants via diffusion along the centerline, and this work allows us to isolate and quantify the effects of diffusion and reaction parameters on the heterogeneous reaction under flowing conditions ($\text{Pe} = \sim 100$).

3. Model Description

[13] The model consists of three components: a flow module, a multicomponent reactive transport module, and a precipitation and dissolution module. We assume that the flow field reaches steady state instantaneously at each time step because the residence time in the numerical domain is short. We also assume that the geometry of the solid phase is constant during each time step when reactive transport occurs, until the reaction alters the liquid phase (or solid phase) in any grid cell.

3.1. Flow Field

[14] The lattice-Boltzmann method (LBM) is used to solve the steady state Stokes equation for interstitial pore velocities as described in our previous work [Knutson *et al.*, 2005]. A standard single-relaxation time model (or Bhatnagar-Gross-Krook [BGK]) [e.g., Hou *et al.*, 1995; Noble *et al.*, 1995] was used. No-slip boundary conditions are imposed at all solid surfaces (i.e., original cylinder posts in the micromodel and precipitated minerals) via a bounce-back boundary condition. Along the horizontal boundaries in the 2-D domain, no-slip boundary is imposed. Fixed pressure boundary conditions are imposed along the inlet ($x = 0$) and outlet ($x = L$) and the resulting steady flow field is scaled to the Darcy velocity used in the experiment. A new steady flow field is solved each time the pore structure changes due to precipitation and dissolution.

3.2. Reactive Transport

[15] We solve the steady state advection-diffusion reaction equation using the velocity field from the LBM. We assume solute concentrations do not change over the time interval between updates of the volumetric fraction of precipitates. We assume that homogeneous reactions are instantaneous, and that all heterogeneous reactions at the mineral surface are kinetically controlled [Lichtner, 1996; Steefel *et al.*, 2005; Kang *et al.*, 2006]. The heterogeneous reaction is treated through the boundary condition on the reactive surface (Γ_s). The reactive transport equations in terms of the total concentration of the j -th primary species (Ψ_j) in the liquid phase (Ω_p) and heterogeneous reaction on the reactive surface (Γ_s) are given by

$$\nabla \cdot (\mathbf{u}\Psi_j - D\nabla\Psi_j) = 0 \quad \text{in } \Omega_p, \quad (1)$$

$$D \frac{\partial \Psi_j}{\partial \mathbf{n}} = -I_m \quad \text{on } \Gamma_s, \quad (2)$$

$$\Psi_j = C_j + \sum_{i=1}^{N_{eq}} \nu_{ji} C_i, \quad (3)$$

where $\mathbf{u}$ is the steady velocity field from LBM, D is the diffusion coefficient, ν_{ji} are the stoichiometric coefficients, $\mathbf{n}$ is the outer unit normal to the interface, I_m is the precipitation-dissolution rate, C_j and C_i are solute concentrations of primary and secondary species in the liquid phase, respectively, and N_{eq} is the number of secondary species (subject to equilibrium reaction). In this study, the same diffusion coefficient for all species is used to avoid the violation of charge balance [Lichtner, 1985; Lichtner and Kang, 2007; Li *et al.*, 2007]. A finite volume method (FVM) with a second-order approximation for advection and diffusion is used to numerically solve the system, and the resulting equations with the heterogeneous reaction (equation (2)) are linearized by the enhanced sequential iterative approach (SIA-1) [Tebe-Stevens *et al.*, 1998]. Fixed concentrations and a zero gradient boundary conditions were used at the inlet and outlet, respectively.

3.3. Reaction Kinetics

[16] Three primary species, Ca^{2+} , H^+ , and CO_3^{2-} , and three secondary species, OH^- , HCO_3^- , and $\text{H}_2\text{CO}_{3(\text{aq})}$ (where $\text{H}_2\text{CO}_{3(\text{aq})}$ represents the sum of H_2CO_3 and $\text{CO}_{2(\text{aq})}$) are considered to model CaCO_3 precipitation and dissolution. As mentioned previously, only the formation of vaterite is considered. The secondary species concentrations (C_i) are given through the mass action law as functions of the primary species concentrations:

$$C_i = \gamma_i^{-1} K_i \prod_{j=1}^{N_c} (\gamma_j C_j)^{\nu_{ji}}, \quad (4)$$

where γ_i is the activity coefficient and K_i is the equilibrium constant. The activity coefficient is computed from the extended Debye-Hückel (D-H) equation:

$$\log \gamma_i = -\frac{Az_i^2 \sqrt{I}}{1 + aB\sqrt{I}}, \quad (5)$$

where A and B are constant for water at 25°C , a is the ion size parameter, z is the valence, and I is the ionic strength. All constant values are taken from Helgeson and Kirkham [1974]. The ionic strength is defined as

$$I = \frac{1}{2} \sum_i m_i z_i^2, \quad (6)$$

where m_i is the molality (where $C_i = \rho m_i$). The nonreactive species (Na^+ and Cl^-) concentrations are solved independently to compute activity coefficients. The system is fully saturated with water (i.e., closed to the atmosphere), and the density of the aqueous phase is equal to pure water (i.e., dilute solution).

[17] The CaCO_3 precipitation-dissolution rate I_m has the form [Chou *et al.*, 1989]:

$$I_m = -k_{cc} (1 - \Omega) = -(k_1 a_{\text{H}^+} + k_2 a_{\text{H}_2\text{CO}_3} + k_3) \left(1 - \frac{Q_{cc}}{K_{sp}}\right), \quad (7)$$

where k_{cc} is the overall reaction rate constant, Ω is the supersaturation index with respect to CaCO_3 , which is defined as the ratio of ion activity product (Q_{cc}) to the saturation product (K_{sp}) of CaCO_3 , k_1 , k_2 , and k_3 are the reaction rate constants, a_i is the activity of species i , and Q_{cc} ($= a_{\text{Ca}^{2+}} a_{\text{CO}_3^{2-}}$) is the ion activity product (IAP) of calcium and carbonate ions. The values of k_1 , k_2 , and k_3 are 8.9×10^{-1} , 5.01×10^{-4} , and $6.6 \times 10^{-7} \text{ mol m}^{-2} \text{ s}^{-1}$, respectively [Chou *et al.*, 1989]. The overall reaction rate constant observed in the work of Chou *et al.* [1989] is relatively large, compared to other values reported in the literature [e.g., Shiraki and Brantley, 1995]. This value is used to account for a highly supersaturated condition in the micromodel experiment, since the growth of calcium carbonate crystals with AFM by Teng *et al.* [2000] was observed to increase with supersaturation index. It is assumed that CaCO_3 precipitation and dissolution occur only at reactive surfaces (equation (2)). The overall reaction rate (R_s) is the product of I_m and the reactive surface area in the grid cell. All equilibrium constants and vaterite properties are presented in Table 1.

3.4. Update of Solid Phase

[18] The solid precipitate phase is assumed immobile, and the volume change of the precipitate is given by

$$\frac{\partial V_m}{\partial t} = \bar{V}_m s_{cc} I_m, \quad (8)$$

where V_m and $\bar{V}_m$ are the volumetric fraction and molar volume of CaCO_3 , and s_{cc} is the specific surface area of the reactive surface for CaCO_3 . Similar approaches have been previously used for multicomponent reactive transport systems [Kang *et al.*, 2006, 2010]. The volumetric fraction of CaCO_3 after each time step is updated explicitly by

$$V_m(t + dt) = V_m(t) + \bar{V}_m s_{cc} I_m \cdot dt. \quad (9)$$

[19] The value of the time interval (dt) is determined as a fraction of the maximum growth or dissolution rate in the system. If the calculated time interval is greater than a

maximum value specified, depending upon the experimental growth rate, the maximum value is used instead in order to avoid a rapid change of the solid phase within one time step. The maximum value of 0.1 min was used at the beginning of the simulation, with a gradual increase to 2 min.

[20] Although the micromodel is considered a 2-D system, precipitate can form on the horizontal top and bottom surfaces as well as on the vertically oriented surfaces of the cylindrical posts. To account for this, each numerical grid cell was conceptualized to have a depth equal to that of the micromodel (i.e., $dz = 20 \text{ } \mu\text{m}$), and the reactive surface area was calculated by assuming CaCO_3 could precipitate both on the bottom and top of the grid cell (i.e., to simulate precipitation on the top and bottom of the micromodel) (as shown in Figure 2) and on vertical walls for grid cells adjacent to a cylindrical post. Mineral precipitates block the pore space, so we use a threshold value for V_m ($V_{m,TV}$), above which it is assumed that there is no flow through a grid cell. Our preliminary numerical simulation results showed that the simulation results with $V_{m,TV}$ below 0.5 or above 0.75 do not agree with experimental results. Within the range from 0.5 to 0.75, the results were only mildly sensitive to $V_{m,TV}$; therefore, $V_{m,TV}$ was set equal to 0.6. The flow field is updated if V_m becomes greater than $V_{m,TV}$ in any grid cell due to mineral reactions after updating the volume of CaCO_3 .

[21] When V_m reaches 1 in a grid cell, the vertical face of the grid cell ($dz \times dx$ or $dz \times dy$) is the reactive surface for precipitation if $\Omega > 1$ in an adjacent cell, or for dissolution if $\Omega < 1$ in an adjacent cell without CaCO_3 (i.e., $V_m = 0$) until V_m is less than $V_{m,TV}$. To account for differences in precipitation and dissolution rates, a dissolution factor (d_f) is introduced as a multiplier in the overall reaction rate; a dissolution factor of 1 means precipitation and dissolution rates are equal. The overall reaction rate (R_s) in a grid cell and volume change of the precipitate are described by

$$R_s = s_{cc} d_f I_m, \quad (10)$$

$$\frac{\partial V_m}{\partial t} = \bar{V}_m s_{cc} d_f I_m, \quad (11)$$

$$V_m(t + dt) = V_m(t) + \bar{V}_m s_{cc} d_f I_m \cdot dt. \quad (12)$$

[22] In the micromodel, d_f is applied after the precipitate along the centerline blocks the pore space, so the region with $V_m < 1$ has $\Omega < 1$. Since the overall reaction rate (equation (10)) is a product of several terms, a value of d_f greater than 1 can correspond to either an increase in the dissolution rate constant or an increase in the reactive surface area. In this work, we are unable to distinguish these two mechanisms, so they are evaluated as a lumped parameter.

[23] Diffusion is reduced in a grid cell occupied by precipitates due to an increase in tortuosity (τ). Hence, we incorporate the effect of τ as V_m changes due to mineral precipitation and dissolution by adapting a form of Archie's law [Archie, 1942]. Archie's law can be expressed as

$$D_{\text{eff}} = D \cdot \tau = D \cdot (1 - V_m)^n, \quad (13)$$

Table 1. Parameter Values Used in Simulations

	Value	SI Units
Diffusion coefficient (D) ^a	9.0×10^{-10}	($\text{m}^2 \text{ s}^{-1}$)
Exponent n	2	—
Reaction rate constant k_1	8.9×10^{-1}	($\text{mol m}^{-2} \text{ s}^{-1}$)
Reaction rate constant k_2	5.01×10^{-4}	($\text{mol m}^{-2} \text{ s}^{-1}$)
Reaction rate constant k_3	6.6×10^{-7}	($\text{mol m}^{-2} \text{ s}^{-1}$)
Threshold for flow, $V_{m,TV}$	0.6	—
Solubility product, K_{sp}	1.832×10^{-2}	($\text{mol}^2 \text{ m}^{-6}$)
Molar volume, $\bar{V}_m$	3.747×10^{-5}	($\text{m}^3 \text{ mol}^{-1}$)

^aFrom Li and Gregory [1974] and Poisson and Papaud [1983].

Reaction rate constants are from Chou *et al.* [1989].

Parameter values are default values unless separately stated.

The total concentrations for each primary species are

$$\Psi_{\text{H}^+} = C_{\text{H}^+} - C_{\text{OH}^-} + C_{\text{HCO}_3^-} + 2C_{\text{H}_2\text{CO}_3}$$

$$\Psi_{\text{CO}_3^{2-}} = C_{\text{CO}_3^{2-}} + C_{\text{HCO}_3^-} + C_{\text{H}_2\text{CO}_3}$$

$$\Psi_j = C_j \text{ for } j = \text{Ca}^{2+}, \text{Na}^+, \text{Cl}^-.$$

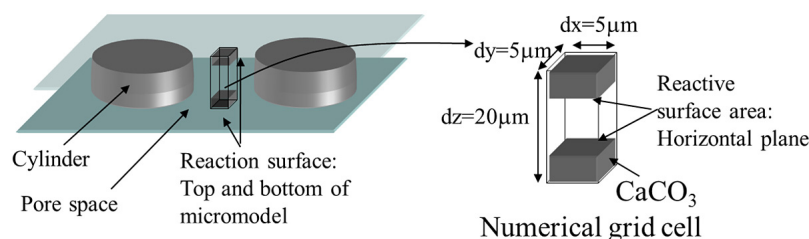


Figure 2. Schematic of reaction model. A numerical grid cell has a dimension of $5 \times 5 \times 20 \mu\text{m}^3$. The horizontal grid area is initially used as the reactive surface area and the reactive surface area evolves as a function of the volumetric content of calcium carbonate as shown in gray blocks (right).

where n represents the cementation exponent in natural materials [Navarre-Sitchler *et al.*, 2009]. In this study, the exponent n was varied from 0 to 3 in order to test the effect of the diffusion coefficient on reaction at the pore scale. In natural materials, the value for n was observed to range from 2 to 2.2 [Boving and Grathwohl, 2001; Navarre-Sitchler *et al.*, 2009].

3.5. Model Application Scenarios

[24] A uniform grid spacing of $5 \mu\text{m}$ is used for both the LB and FV methods, which results in 60 grid cells across a grain and seven grid cells across a pore throat. Additional simulation results of precipitate in the first pore space with a $1 \mu\text{m}$ grid were not significantly different (i.e., $<3\%$ in precipitate mass) from the results with a $5 \mu\text{m}$ grid; hence, only the results with a $5 \mu\text{m}$ grid are presented in this study. As mentioned, the depth of the numerical domain (i.e., one grid in the vertical direction) is $20 \mu\text{m}$. The grid spacing of $5 \mu\text{m}$ has previously been used to simulate different biochemical reactions in an identical pore structure [Knutson *et al.*, 2005; Willingham *et al.*, 2008, 2010].

[25] Representative chemical and physical properties used in the simulations are listed in Table 1. Inlet concentrations for total concentrations of calcium, carbonate, and hydrogen ions are computed based upon the experimental values of the inlet pH and concentrations of CaCl_2 and Na_2CO_3 , equilibrium constants, and activity coefficients computed using equation (5). The total carbonate concentration in CaCl_2 solution was assumed negligible ($\ll 10^{-5} \text{ mol L}^{-1}$) compared to that in a Na_2CO_3 solution because the former solution was degassed in a vacuum. As mentioned earlier, the diffusion coefficient of all aqueous species is the same and set to $0.9 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$ [Li and Gregory, 1974; Poisson and Papaud, 1983].

[26] Five separate cases with different parameter sets were used to test the impact of the following key parameters: effective diffusion coefficient, overall reaction rate, and dissolution factor. There are summarized in Table 2. Case 1 is considered the base case. For this case, the exponent n for D_{eff} was set equal to 2, reaction kinetic constants are given by Chou *et al.* [1989], and the dissolution factor was set to 1. The exponent n was changed to 0 and 3 for cases 2 and 3, respectively. When n is equal to zero the diffusion coefficient is constant and not affected by pore blockage due to precipitation. Case 4 has different k_1 , k_2 , and k_3 values, each equal to twice the base case values in case 1. For case 5, the dissolution factor is changed to better match the data.

4. Results

4.1. Base Case (Case 1)

[27] Images of CaCO_3 precipitates in the micromodel and corresponding simulation results at 13, 18, and 118 min are shown in Figure 3. The measured and simulated precipitate areas in the entire numerical domain (i.e., the first three pore spaces) are also shown in Figure 4a. The simulated precipitate area was determined for grid cells with a value of V_m greater than 0.05. Other threshold values were evaluated, but resulted in a poorer fit of the experimental data using base case parameter values. Transverse mixing between calcium and carbonate ions created a supersaturated zone along the central mixing line, yielding simulated Ω values in the first pore body of ~ 2100 before a reaction occurred and up to ~ 800 after a reaction occurred (Figure 4b). This resulted in rapid precipitation in the first 13 min. In each pore body, more diffusive mixing occurred with distance, resulting in a wider precipitate zone downstream than upstream. Experimental images at 13 and 18 min show that the precipitation area in the first pore space was quite similar, but the darkness of the precipitate zone increased, suggesting that V_m increased over time along the centerline. The precipitation patterns in the other pore bodies followed the pattern in the first pore body. Precipitation was followed by dissolution over ~ 40 min (Figures 3c and 4a). Eventually, the precipitate area reached an asymptotic level with a very thin precipitate line along the centerline of each pore body (Figure 3c), which lasted over 3 h until the end of the experiment. Experimental images suggest that as V_m increases, diffusive mixing of Ca^{2+} and CO_3^{2-} was reduced, and this led to dissolution along the outer edges of the precipitate zone.

[28] Simulation results in Figures 3 and 4 appear to capture both the initial rate of precipitation and the precipitation

Table 2. Conditions Used in Simulation Cases

Case	Exponent Constant, n	Kinetic Rate ^a ($\text{mol cm}^{-2} \text{ s}^{-1}$)	Dissolution Factor
1 ^b	2	k_{cc}	1
2	0	k_{cc}	1
3	3	k_{cc}	1
4	2	$k_{cc} \times 2$	1
5	2	k_{cc}	300

^aThe default values are from Chou *et al.* [1989]. k_{cc} is the overall reaction rate constant in equation (7).

^bCase 1 is the base case. Other cases use conditions defined in case 1 except where conditions are described.

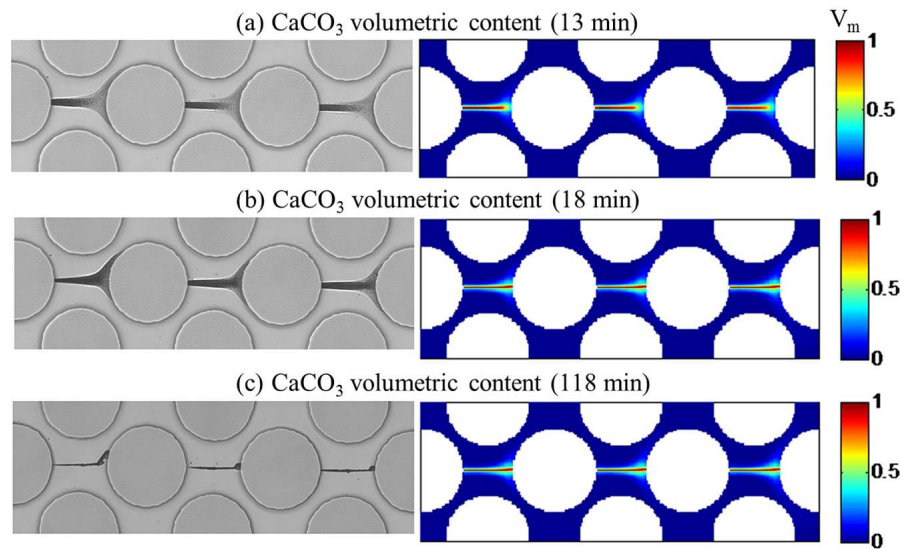


Figure 3. Image of CaCO_3 precipitates and simulated CaCO_3 volumetric content at (a) 13 min, (b) 18 min, and (c) 118 min, for case 1.

morphology observed in the micromodel. Before pore blocking, the pH increased dramatically in the mixing zone due to reaction (i.e., carbonate ion depletion) and diffusion (Figure 4c). This indicates that the effect of pH on the precipitation rate is likely to be insignificant as long as diffusion is allowed across the centerline. However, simulations were not able to capture dissolution after ~ 15 min. After 24 min, precipitates were sufficiently dense across each pore body, such that Ω became < 1 off the centerline (i.e., undersaturated conditions). However, simulated dissolution is much slower compared to the experimental results (Figure 4a), indicating that less diffusive mixing after pore blockage, greater kinetic rate constants, and/or a higher reactive surface area during dissolution (equations (10)–(12)) are required.

4.2. Effective Diffusion Coefficient (Cases 1–3)

[29] The influence of the effective diffusion coefficient, i.e., the parameter n (see equation (13)), on the amount of precipitate over time, and the corresponding experimental

image after 13 min are shown in Figure 5. The measured and simulated precipitate areas are also shown (Figure 5f). For the default case of $n = 2$ (case 1), D_{eff} decreases as V_m increases and this limits the amount of precipitate and yields good agreement with the experimental data up to ~ 13 min. As the n -value decreases, D_{eff} decreases less as V_m increases, resulting in more precipitation (i.e., a higher-precipitation rate) compared to the experimental results. As the n -value increases, D_{eff} decreases more as V_m increases, resulting in less precipitation due to a smaller reaction zone. However, the range of n -values evaluated does not adequately capture the decrease in the precipitate area that occurs after ~ 13 min.

4.3. Overall Reaction Constant (Cases 1 and 4)

[30] The influence of overall reaction constant (k_{cc}) on precipitation and dissolution and a corresponding experimental image after 13 min are shown in Figures 5d and 5e. The measured and simulated precipitate areas are also shown (Figure 5f). The k_{cc} value for case 4 is multiplied

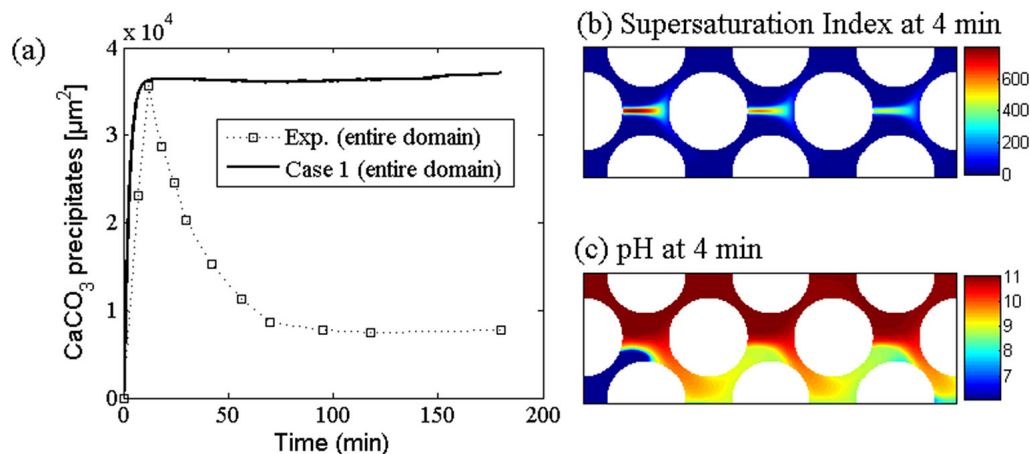


Figure 4. (a) Comparison of CaCO_3 precipitates area in the entire numerical domain, (b) supersaturation index at 4 min, and (c) pH at 4 min (case 1).

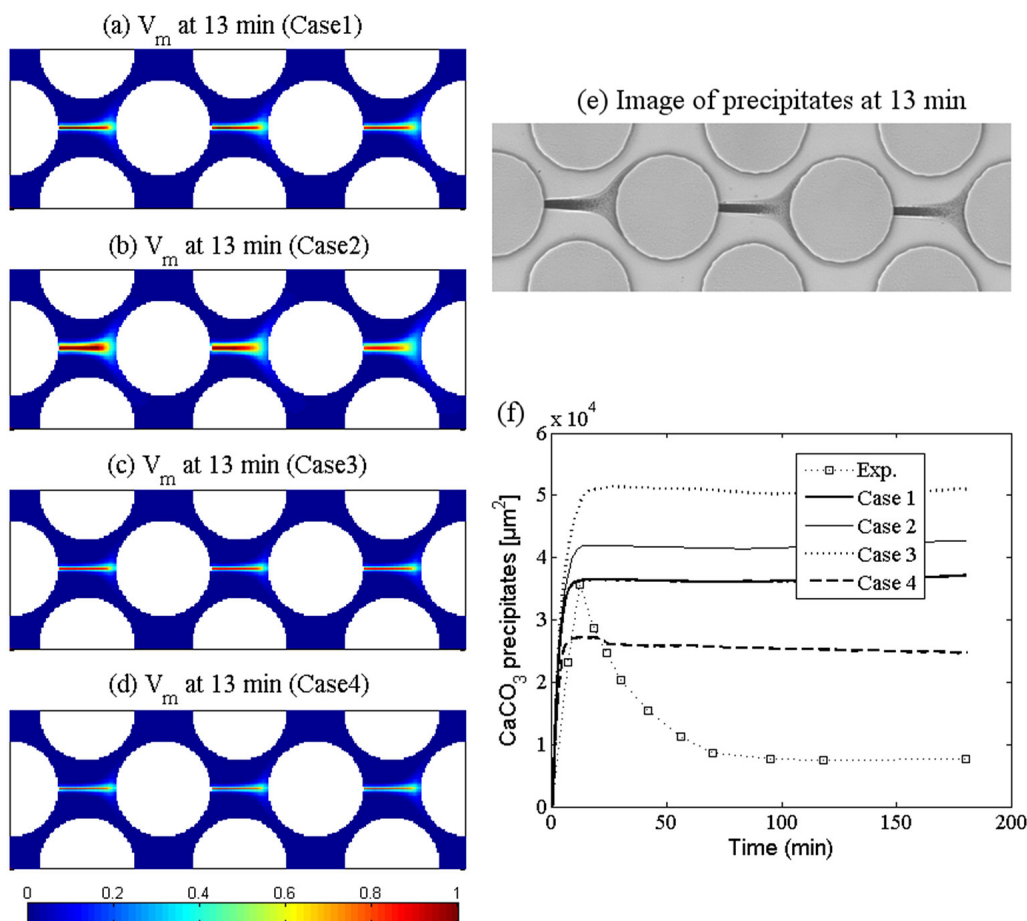


Figure 5. Effect of the exponent n and overall reaction constant on CaCO_3 volumetric content (V_m). (a) $n = 2$ (case 1), (b) $n = 0$ (case 2), (c) $n = 3$ (case 3), and (d) $k_{cc} \times 2$ (case 4). (e) Experimental image of precipitates at 13 min, and (f) measured and simulated precipitate areas over time.

by a factor of two, compared to that for case 1. As k_{cc} increases, precipitation occurs faster, particularly along the centerline, resulting in a reduction of diffusion (i.e., mixing). Hence, increasing k_{cc} results in decreasing the maximum precipitate area. After the maximum precipitation area is reached, some dissolution is predicted but at a much slower rate than the experimental data. Hence, the reaction rate from Chou *et al.* [1989] appears to adequately capture the growth rate at early time, as well as the maximum area of precipitation for the calibrated threshold value of $V_m = 0.05$, but the extent of dissolution is not adequately captured.

4.4. Dissolution Factor (Case 5)

[31] Dissolution factors (d_f) ranging from 10 to 1000 were evaluated and the results for d_f greater than 300 were nearly identical. The simulation results with d_f equal to 300 best matched the experimental data, and are presented with experimental images at three different times in Figure 6. The measured and simulated precipitate areas are shown in Figure 7a. In contrast to case 1, the thickness and area of precipitate along the centerline after 13 min for case 5 decreases more quickly, and the precipitate area matches the experimental data until 30 min. In addition, as in the

experimental data, the model predicts that dissolution below the centerline occurs more quickly than dissolution above the centerline. This is because of the higher pH above the centerline after pore blocking (Figures 7b and 7c), resulting in a higher supersaturation index above the centerline (Figure 7d). Hence, the discrepancy in the precipitate area between the simulation and the experimental results after 30 min is attributed to slow dissolution above the centerline.

5. Summary and Conclusions

[32] A new 2-D pore-scale reactive transport model was used to explore the mechanisms of transverse-mixing induced CaCO_3 precipitation and dissolution in a micromodel experiment [Zhang *et al.*, 2010]. The model considers the effects of pH on carbonate speciation and a CaCO_3 reaction rate constant, of changing reactive surface area on the reaction, and of pore blockage from CaCO_3 precipitation on diffusion and flow. The reactive surface area during precipitation in a grid cell was assumed to include the top and bottom surfaces of the micromodel and the cylinder walls for grid cells adjacent to 2-D grains. In the experiment, the precipitate area increased rapidly due to fast precipitation for ~ 13 min, decreased rapidly due to fast dissolution within 30 min, and then approached a relatively

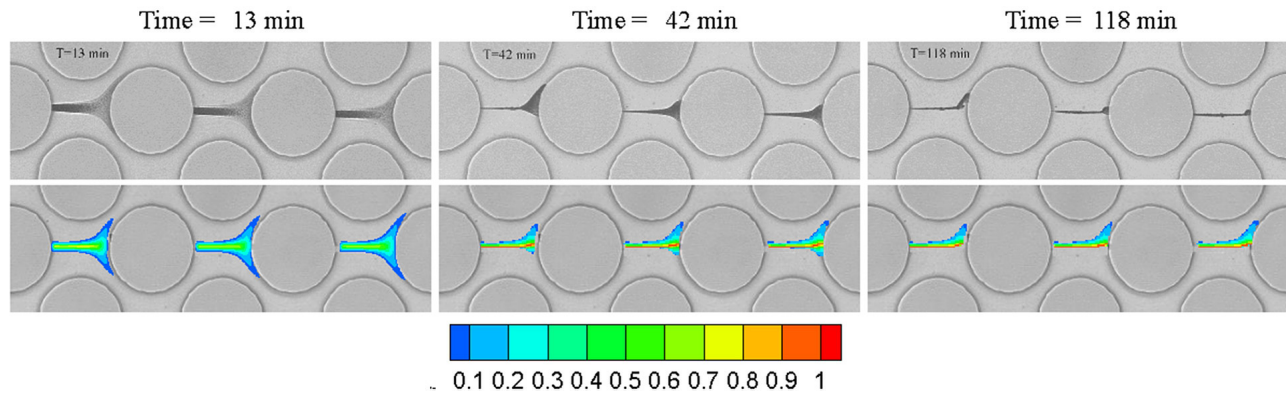


Figure 6. Image of CaCO_3 precipitates (top) and simulated CaCO_3 volumetric content (bottom) at three different times (case 5). A dissolution factor of 300 is used for case 5.

constant but lower plateau over 3–4 h. In simulation results, the grid cells with $V_m > 0.05$ were counted as precipitate. Overall, the pore scale model using parameter values from the literature captured the precipitate morphology, precipitation rate, and maximum precipitation area from the experiment. However, a dissolution factor was required to simulate the rapid decrease in precipitate area after 13 min, and this only allowed the model to capture the precipitate area until 30 min. After this time, the simulated precipitate area leveled off, while the experimental precipitate area continued to decrease.

[33] The effective diffusion parameter (D_{eff}) that accounted for the effect of precipitate on hindered diffusion was expressed as a function of V_m (equation (13)). Greater n -values resulted in slower diffusion with increasing V_m , and less overall precipitate in the pore space. However, values of n between 0 and 3 did not improve the simulation of the rapid dissolution observed in the experiment after

13 min. The value of the overall reaction rate constant, k_{cc} , was increased two times to explore the effects on CaCO_3 precipitation and dissolution. A higher k_{cc} value increased the overall rate of precipitation along the centerline during the first 13 min (equation (7)), and decreased the maximum amount of precipitate due to fast pore blocking, resulting in a poorer fit of the data. However, changing k_{cc} did not improve the simulation of the rapid dissolution observed in the experiment after 13 min.

[34] The need to include a dissolution factor to capture the rapid decrease in the precipitate area after 13 min may be attributed to an increase in the precipitate surface area and the recrystallization process during dissolution. We initially assume the precipitate surface area is equal to the surface area of the micromodel top and bottom and the cylinder walls when a grid cell is adjacent to a 2-D grain. It is somewhat surprising that this simple assumption allows us to capture the precipitation rate during the first 13 min.

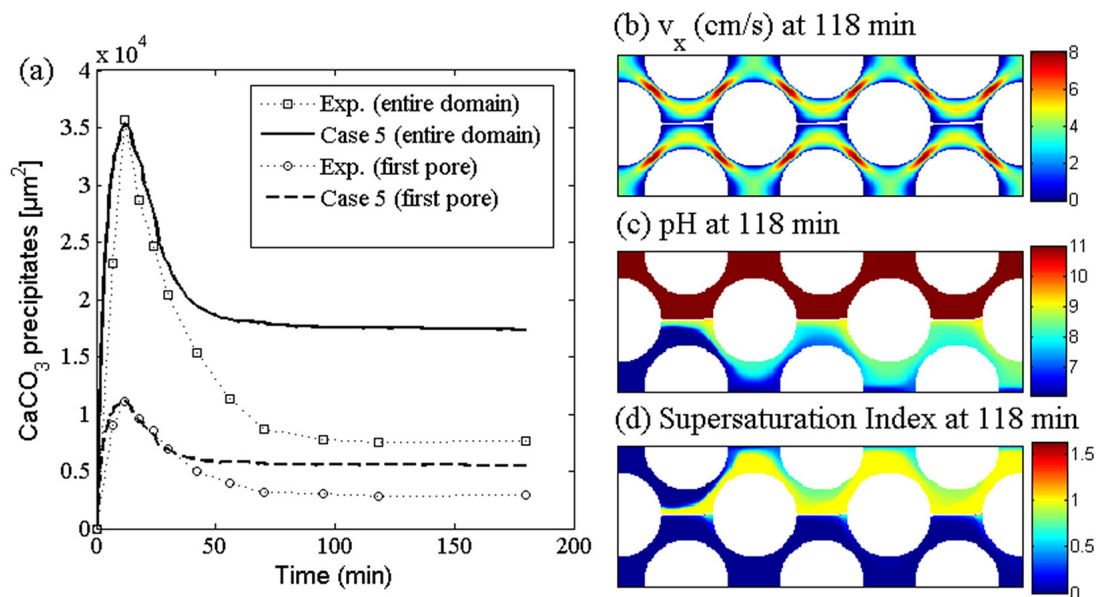


Figure 7. (a) Comparison of CaCO_3 precipitates area in the first pore space (case 5) and the entire numerical domain (case 5), (b) horizontal velocity, (c) pH, and (d) supersaturation index at 118 min (case 5). White color in pore bodies in (b) corresponds to grid cells with $V_m > 0.95$.

However, it is not evident why a different precipitation and dissolution rate are required to match the overshoot until 30 min. Shao *et al.* [2010] observed nanometer-scale pitting on a phlogopite surface during dissolution with supercritical CO₂ using atomic force microscopy. In an earlier reactive flow model, an increase in the reactive surface area over time was considered by using crystal-size distributions that change due to heterogeneous nucleation and the effect of the Ostwald ripening on crystal growth [Steeff and Van Cappellen, 1990]. Rieger *et al.* [2000] used transmission X-ray microscopy to observe that freshly formed vaterite crystals at the micron scale can dissolve completely within 6 min and recrystallize into bigger particles by diffusion. More recently, Pouget *et al.* [2009] used cryotransmission electron microscopy and observed nanoparticles with a diameter of ~30 nm adhere to the surface of vaterite crystals, and the growth of the vaterite particles occurs through a dissolution-reprecipitation mechanism of nanoparticles. While these studies did not measure the reactive surface area, it is possible that such pitting and the presence of nanoparticles increases the reactive surface area and enhances dissolution under flowing conditions.

[35] Our model allows us to simulate and mechanistically evaluate the precipitation and dissolution of CaCO₃ that was observed in a micromodel pore network. The overall rate and amount of precipitation are reasonably captured and mechanistically explained, but not the enhanced dissolution that occurs due to pore blockage and reduced mixing. We note that the reactive surface area may increase during dissolution due to nanometer scale features such as pitting and the presence of nanoparticles. However, direct evidence of this process in our system is lacking. Further work is needed to mechanistically evaluate the reason for enhanced dissolution in the micromodel experiment. This has implications for geological carbon sequestration, because the extent of precipitation at all times during CO₂ injection may affect pore blockage, the efficiency of CO₂ injection, and the ability to mix and sequester CO₂ in the geologic formation.

[36] **Acknowledgments.** H.Y. and T. D. were supported as part of the Center for Frontiers of Subsurface Energy Security, an Energy Frontier Research Center funded by the U.S. Department of Energy, Office of Science, Office of Basic Energy Sciences under Award DE-SC0001114. Sandia National Laboratories is a multiprogram laboratory managed and operated by Sandia Corporation, a wholly owned subsidiary of Lockheed Martin Corporation, for the U.S. Department of Energy's National Nuclear Security Administration under contract DE-AC04-94AL85000. Additional support for A.J.V. and C.J.W. was provided by LDRD 82552-001-10 sponsored by the Los Alamos National Laboratory. We also acknowledge the effort of Carl I. Steeff, Alexander M. Tartakovsky, and Qinjin Kang for their careful and constructive reviews, which led to improvement of our manuscript.

References

- Archie, G. E. (1942), The electrical resistivity log as an aid in determining some reservoir characteristics, *Trans. Soc. Pet. Eng.*, **146**, 54–62.
- Balhoff, M. T., S. G. Thomas, and M. F. Wheeler (2008), Mortar coupling and upscaling of pore-scale models, *Comput. Geosci.*, **12**(1), 15–27.
- Battiato, I., D. M. Tartakovsky, A. M. Tartakovsky, and T. D. Scheibe (2011), Hybrid simulations of reactive transport in porous and fractured media, *Adv. Water Res.*, **34**(9), 1140–1150, doi:10.1016/j.advwatres.2011.01.012.
- Blum, A. E., and A. C. Lasaga (1987), Monte Carlo simulations of surface reaction rate laws, in *Aquatic Surface Chemistry: Chemical Processes at the Particle-Water Interface*, edited by W. Stumm, pp. 255–291, Wiley, N. Y.
- Boving, T. B., and P. Grathwohl (2001), Tracer diffusion coefficients in sedimentary rocks: Correlation to porosity and hydraulic conductivity, *J. Contam. Hydrol.*, **53**, 85–100.
- Chomsurin, C., and C. J. Werth (2003), Analysis of pore-scale nonaqueous phase liquid dissolution in etched silicon pore networks, *Water Resour. Res.*, **39**(9), 1265, doi:10.1029/2002WR001643.
- Chou, L., R. M. Garrels, and R. Wollast (1989), Comparative study of the kinetics and mechanisms of dissolution of carbonate minerals, *Chem. Geol.*, **78**, 269–282.
- Dehoff, K. (2010), Pore-scale characterization of transverse mixing-induced calcium carbonate precipitation of relevance to geological carbon sequestration, M.S. Thesis, University of Illinois, Urbana, IL.
- De Yoreo, J. J., and P. Vekilov (2003), Principles of crystal nucleation and growth, in *Biomimetic Mineralization*, edited by P. M. Dove, J. J. De Yoreo, and S. Weiner, pp. 57–93, Mineralogical Society of America, Washington, D.C.
- De Yoreo, J. J., L. A. Zepeda-Ruiz, R. W. Friddle, G. H. Gilmer, A. A. Chernov, and P. M. Dove (2009), Rethinking classical crystal growth models through molecular scale insights: Consequences of kink-limited kinetics, *J. Cryst. Growth*, **9**, 5135–5144, doi:10.1021/cg900543g.
- Emmanuel, S., and B. Berkowitz (2005), Mixing-induced precipitation and porosity evolution in porous media, *Adv. Water Res.*, **28**, 337–344.
- Flukiger, F., and D. Bernard (2009), A new numerical model for pore scale dissolution of calcite due to CO₂ saturated water flow in 3D realistic geometry: Principles and first results, *Chem. Geol.*, **265**(1–2), 171–180.
- Gouze, P., C. Noiriell, C. Bruderer, D. Loggia, and R. Leprovost (2003), X-ray tomography characterization of fracture surfaces during dissolution, *Geophys. Res. Lett.*, **30**(5), 1267, doi:10.1029/2002GL016755.
- Helgeson, H. C., and D. H. Kirkham (1974), Theoretical prediction of the thermodynamic behavior of aqueous electrolytes at high pressures and temperatures. II. Debye-Hückel parameters for activity coefficients and relative partial molal properties, *Am. J. Sci.*, **274**(10), 1199–1261.
- Hostomsky, J., and A. G. Jones (1991), Calcium carbonate crystallization, agglomeration and form during continuous precipitation from solution, *J. Phys. D Appl. Phys.*, **24**, 165–170.
- Hou, S., Q. Zou, S. Chen, G. Doolen, and A. C. Cogley (1995), Simulations of cavity flow by the lattice Boltzmann method, *J. Comput. Phys.*, **118**, 329–347.
- Inskeep, W. P., and P. R. Bloom (1985), An evaluation of rate equations for calcite precipitation kinetics at pCO₂ less than 0.01 atm and pH greater than 8, *Geochim. Cosmochim. Acta*, **49**, 2165–2180.
- Johnson, J. W., J. J. Nitao, and K. G. Knauss (2004), Reactive transport modeling of CO₂ storage in saline aquifers to elucidate fundamental processes, trapping mechanisms, and sequestration partitioning, in *Geological Storage of Carbon Dioxide*, edited by S. J. Baines and R. H. Worden, pp. 107–128, Geological Society of London, London, U. K.
- Kang, Q., D. Zhang, and S. Chen (2003), Simulation of dissolution and precipitation in porous media, *J. Geophys. Res.*, **108**(B10), 2505, doi:10.1029/2003JB002504.
- Kang, Q., P. C. Lichtner, and D. Zhang (2006), Lattice Boltzmann pore-scale model for multicomponent reactive transport in porous media, *J. Geophys. Res.*, **111**, B05203, doi:10.1029/2005JB003951.
- Kang, Q., P. C. Lichtner, and D. Zhang (2007), An improved lattice Boltzmann model for multicomponent reactive transport in porous media at the pore scale, *Water Resour. Res.*, **43**, W12S14, doi:10.1029/2006WR005551.
- Kang, Q., P. C. Lichtner, H. S. Viswanathan, and A. I. Abdel-Fattah (2010), Pore scale modeling of reactive transport involved in geologic CO₂ sequestration, *Transp. Porous Media*, **82**(1), 197–213.
- Kellermeier, M., E. Melero-García, F. Glaab, R. Klein, M. Drechsler, R. Rachel, J. M. García-Ruiz, and W. Kunz (2010), Stabilization of amorphous calcium carbonate in inorganic silica-rich environments, *J. Am. Chem. Soc.*, **132**(50), 17859–17866.
- Kim, D., W. B. Lindquist, and C. A. Peters (2011), Upscaling geochemical reaction rates accompanying acidic CO₂-saturated brine flow in sandstone aquifers, *Water Resour. Res.*, **47**, W01505, doi:10.1029/2010WR009472.
- Knutson, C., C. J. Werth, and A. J. Valocchi (2005), Pore scale simulations of biomass growth along the transverse mixing zone of a model two-dimensional porous medium, *Water Resour. Res.*, **41**, W07007, doi:10.1029/2004WR003459.
- Li, L., C. A. Peters, and M. A. Celia (2006), Upscaling geochemical reaction rates using pore-scale network modeling, *Adv. Water Res.*, **29**, 1351–1370.

- Li, L., C. A. Peters, and M. A. Celia (2007), Reply to "Comment on upscaling geochemical reaction rates using pore-scale network modeling" by P. C. Lichtner and Q. Kang, *Adv. Water Res.*, **30**, 691–695.
- Li, Y. H., and S. Gregory (1974), Diffusion of ions in seawater and in deep sea sediments, *Geochim. Cosmochim. Acta*, **38**, 703–714.
- Lichtner, P. C. (1985), Continuum model for simultaneous chemical reactions and mass transport in hydrothermal systems, *Geochim. Cosmochim. Acta*, **49**(3), 779–800.
- Lichtner, P. C. (1996), Continuum formulation of multicomponent-multiphase reactive transport, in *Reactive Transp. Porous Media*, edited by P. C. Lichtner, C. I. Steefel, and E. H. Oelkers, Reviews in Mineralogy vol. 34, pp. 1–79, Mineralogical Society of America, Washington, D.C.
- Lichtner, P. C., and Q. Kang (2007), Comment on: "Upscaling geochemical reaction rates using pore-scale network modeling" by Li, Peters and Celia, *Adv. Water Res.*, **30**, 686–690.
- Liu, J., S. Pancera, V. Boyko, A. Shukla, T. Narayanan, and K. Huber (2010), Evaluation of the particle growth of amorphous calcium carbonate in water by means of the Porod invariant from SAXS, *Langmuir*, **26**(22), 17405–17412.
- Mozley, P. S., and J. M. Davis (2005), Internal structure and mode of growth of elongate calcite concretions: Evidence for small-scale, microbially induced, chemical heterogeneity in groundwater, *Geol. Soc. Am. Bull.*, **117**, 1400–1412.
- Noiriel, C., P. Gouze, and D. Bernard (2004), Investigation of porosity and permeability effects from microstructure changes during limestone dissolution, *Geophys. Res. Lett.*, **31**, L24603, doi:10.1029/2004GL021572.
- Navarre-Sitchler, A., and S. Brantley (2007), Basalt weathering across scales, *Earth Planet. Sci. Lett.*, **261**(1–2), 321–334.
- Navarre-Sitchler, A., C. I. Steefel, L. Yang, L. Tomutsa, and S. L. Brantley (2009), Evolution of porosity and diffusivity associated with chemical weathering of a basalt clast, *J. Geophys. Res.*, **114**, F02016, doi:10.1029/2008JF001060.
- Noble, D. R., S. Chen, J. G. Gerogiadiis, and R. O. Buckius (1995), A consistent hydrodynamic boundary-condition for the lattice Boltzmann method, *Phys. Fluids*, **7**, 203–209.
- Ogino, T., T. Suzuki, and K. Sawada (1987), The formation and transformation mechanism of calcium carbonate in water, *Geochim. Cosmochim. Acta*, **51**(10), 2757–2767.
- Plummer, L. N., T. M. L. Wigley, and D. L. Parkhurst (1978), The kinetics of calcite dissolution in CO₂-water systems at 5 to 608°C and 0.0 to 1.0 atm CO₂, *Am. J. Sci.*, **278**, 179–216.
- Poisson, A., and A. Papaud (1983), Diffusion coefficients of major ions in seawater, *Mar. Chem.*, **13**(4), 265–280, doi:10.1016/0304-4203(83)90002-6.
- Pontoni, D., J. Bolze, N. Dingenouts, T. Narayanan, and M. Ballauff (2003), Crystallization of calcium carbonate observed in-situ by combined small- and wide-angle X-ray scattering, *J. Phys. Chem. B*, **107**(22), 5123–5125.
- Pouget, E. M., P. H. H. Bomans, J. Goos, P. M. Frederik, G. de With, and N. Sommerdijk (2009), The initial stages of template-controlled CaCO₃ formation revealed by cryo-TEM, *Science*, **323**, 1455–1458.
- Richter, F. H., E. W. Winkler, and R. H. Baur (1989), The calcium binding capacity of polycarboxylates, *J. Am. Oil Chem. Soc.*, **66**(11), 1666–1672.
- Riebesell, U., I. Zondervan, B. Rost, P. D. Tortell, R. E. Zeebe, and F. M. M. Morel (2000), Reduced calcification of marine plankton in response to increased atmospheric CO₂, *Nature*, **407**, 364–367, doi:10.1038/35030078.
- Rieger, J., J. Thieme, and C. Schmidt (2000), Study of precipitation reactions by X-ray microscopy: CaCO₃ precipitation and the effect of polycarboxylates, *Langmuir*, **16**(22), 8300–8305, doi:10.1021/la0004193.
- Rieger, J., T. Frechen, G. Cox, W. Heckmann, C. Schmidt, and J. Thieme (2007), Precursor structures in the crystallization/precipitation processes of CaCO₃ and control of particle formation by polyelectrolytes, *Faraday Discuss.*, **136**, 265–277, doi:10.1039/B701450C.
- Rochelle, C. A., I. Czernichowski-Lauriol, and A. E. Milodowski (2004), The impact of chemical reactions on CO₂ storage in geological formations: A brief review, in *Geological Storage of Carbon Dioxide*, edited by S. J. Baines and R. H. Worden, pp. 87–106, Geological Society of London, London, U. K.
- Shao, H., J. R. Ray, and Y.-S. Jun (2010), Dissolution and precipitation of clay minerals under geologic CO₂ sequestration conditions: CO₂-brine-phlogopite interactions, *Environ. Sci. Technol.*, **44**(15), 5999–6005.
- Shiraki, R., and S. L. Brantley (1995), Kinetics of near-equilibrium calcite precipitation at 100°C: an evaluation of elementary reaction-based and affinity-based rate laws, *Geochim. Cosmochim. Acta*, **59**(8), 1457–1471.
- Stack, A. G., and M. C. Grantham (2010), Growth rate of calcite steps as a function of aqueous calcium-to-carbonate ratio: Independent attachment and detachment of calcium and carbonate ions, *J. Crystal Growth*, **10**, 1409–1413, doi:10.1021/cg901395z.
- Steefel, C. I., and P. C. Lichtner (1994), Diffusion and reaction in rock matrix bordering a hyperalkaline fluid-filled fracture, *Geochim. Cosmochim. Acta*, **58**(17), 3592–3612.
- Steefel, C. I., and P. C. Lichtner (1998), Multicomponent reactive transport in discrete fractures: II. Infiltration of hyperalkaline groundwater at Maqarin, Jordan, a natural analogue site, *J. Hydrol.*, **209**, 200–224.
- Steefel, C. I., and P. Van Cappellen (1990), A new kinetic approach to modeling water-rock interaction: The role of nucleation, precursors, and Ostwald ripening, *Geochim. Cosmochim. Acta*, **54**, 2657–2677.
- Steefel, C. I., D. J. DePaolo, and P. C. Lichtner (2005), Reactive transport modeling: An essential tool and a new research approach for the Earth sciences, *Earth Planet. Sci. Lett.*, **240**, 539–558.
- Tartakovsky, A. M., and T. D. Scheibe (2011), Dimension reduction method for advection-diffusion-reaction systems, *Adv. Water Res.*, **34**(12), 1616–1626, doi:10.1016/j.advwatres.2011.07.011:2011.
- Tartakovsky, A. M., P. Meakin, T. D. Scheibe, and B. D. Wood (2007), A smoothed particle hydrodynamics model for reactive transport and mineral precipitation in porous and fractured porous media, *Water Resour. Res.*, **43**(5), W05437, doi:10.1029/2005WR004770.
- Tartakovsky, A. M., G. Redden, P. C. Lichtner, T. D. Scheibe, and P. Meakin (2008), Mixing-induced precipitation: Experimental study and multiscale numerical analysis, *Water Resour. Res.*, **44**, W06S04, doi:10.1029/2006WR005725.
- Tartakovsky, A. M., T. D. Scheibe, and P. Meakin (2009), Pore-scale model for reactive transport and biomass growth, *J. Porous Media*, **12**(5), 417–434.
- Tebes-Stevens, C., A. J. Valocchi, J. M. VanBriesen, and B. E. Rittmann (1998), Multicomponent transport with coupled geochemical and microbiological reactions: Model description and example simulations, *J. Hydro.*, **209**(1–4), 8–26, doi:10.1016/S0022-1694(98)00104-8.
- Teng, H. H., P. M. Dove, and J. J. DeYoreo (2000), Kinetics of calcite growth: Surface processes and relationships to macroscopic rate laws, *Geochim. Cosmochim. Acta*, **64**, 2255–2266.
- Whitaker, S. (1999), *The Method of Volume Averaging*, 219 pp., Kluwer, Dordrecht, Netherlands.
- Willingham, T. W., C. J. Werth, and A. J. Valocchi (2008), Evaluation of the effects of porous media structure on mixing-controlled reactions using pore-scale modeling and micromodel experiments, *Environ. Sci. Technol.*, **42**(9), 3185–3193.
- Willingham, T., C. Zhang, C. J. Werth, A. J. Valocchi, M. Oostrom, and T. W. Wietsma (2010), Using dispersivity values to quantify the effects of pore-scale flow focusing on enhanced reaction along transverse mixing zone, *Adv. Water Res.*, **33**, 525–535.
- Wood, B. D. (2009), The role of scaling laws in upscaling, *Adv. Water Res.*, **32**(5), 723–736.
- Zhang, C., K. Dehoff, N. Hess, M. Oostrom, T. W. Wietsma, A. J. Valocchi, B. W. Fouke, and C. J. Werth (2010), Pore-scale study of transverse mixing induced CaCO₃ precipitation and permeability reduction in a model subsurface sedimentary system, *Environ. Sci. Technol.*, **44**(20), 7833–7838.

T. Dewers and H. Yoon, Geomechanics Department, Sandia National Laboratories, P.O. Box 5800, MS 0751, Albuquerque, NM 87185, USA. (hyoon@sandia.gov)

A. J. Valocchi and C. J. Werth, Department of Civil and Environmental Engineering, University of Illinois at Urbana-Champaign, 205 N. Mathews Ave. (MC-250), Urbana, IL 61801, USA.